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(54) **COMPOSITION FOR PREVENTING OR TREATING DEGENERATIVE BRAIN DISEASES**

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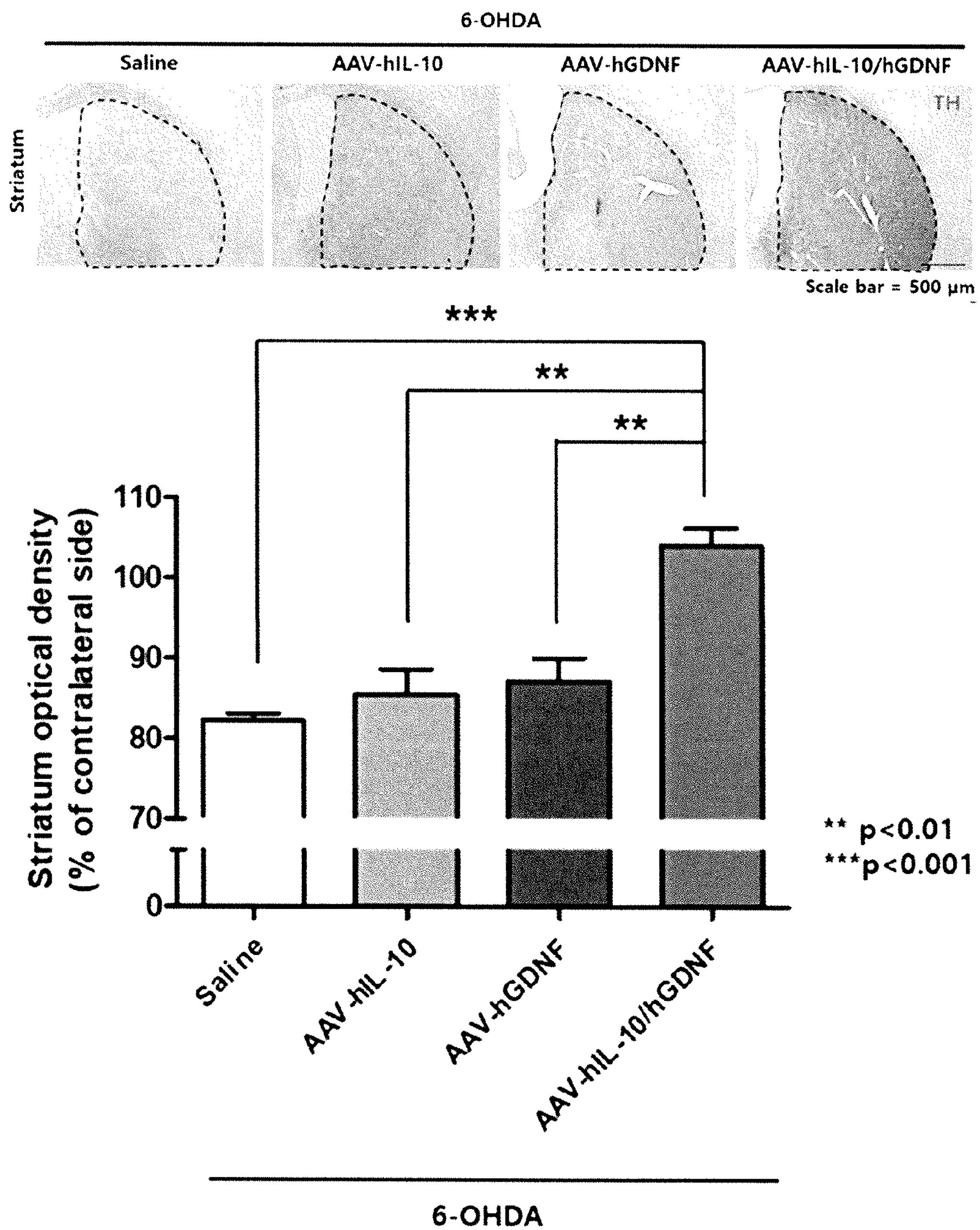
(57) **ABSTRACT**

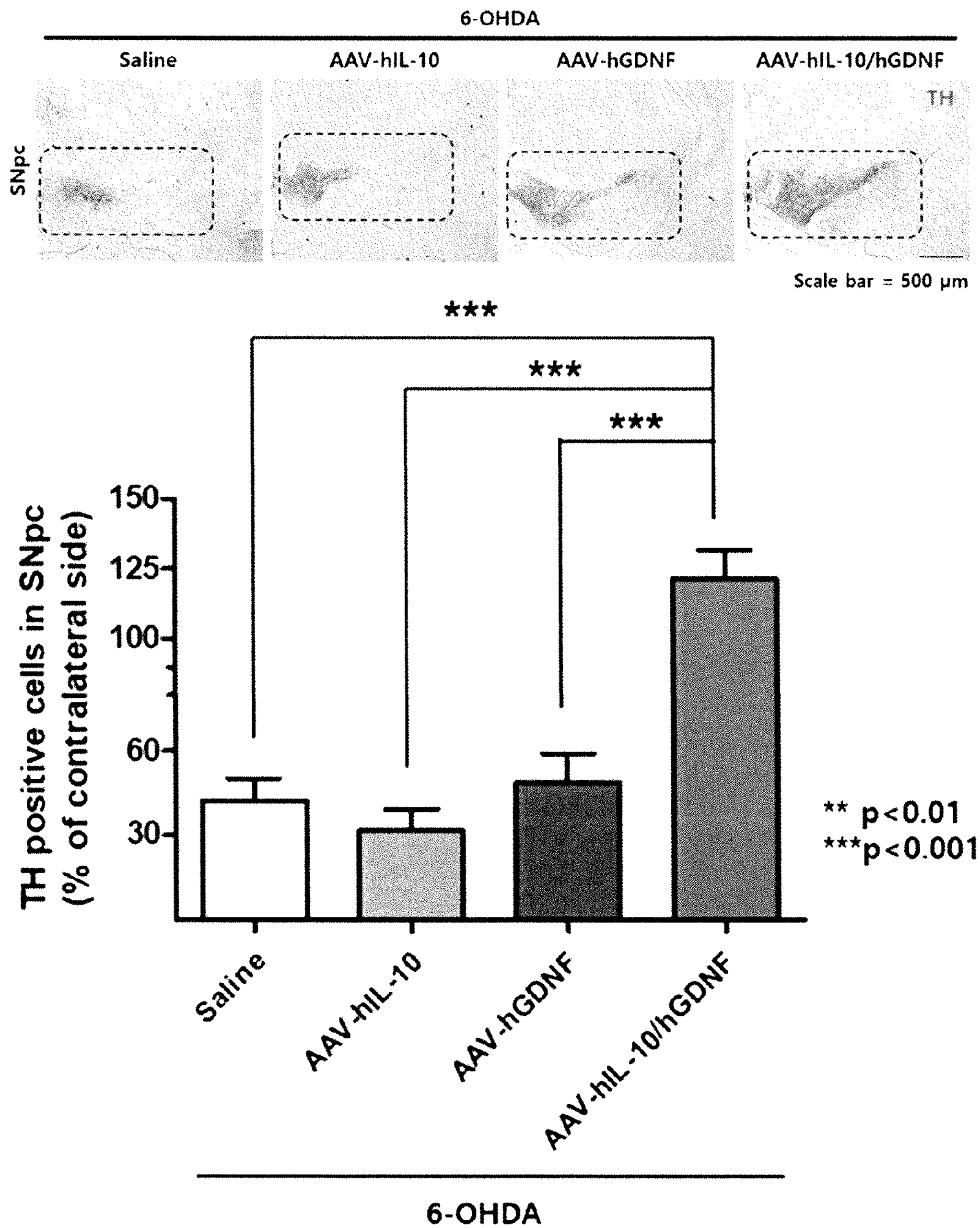
The present invention relates to a composition for preventing or treating neurodegenerative diseases. The pharmaceutical composition of the present invention contains at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same. Since the pharmaceutical composition of the present invention exhibits a better effect of alleviating and preventing neurodegenerative diseases by co-administration of active ingredients than upon single administration, the pharmaceutical composition of the present invention can exhibit the same or higher effect only with a low concentration of the active ingredients. Therefore, the pharmaceutical composition of the present invention can be advantageously used for the prevention or treatment of neurodegenerative diseases.

Specification includes a Sequence Listing.

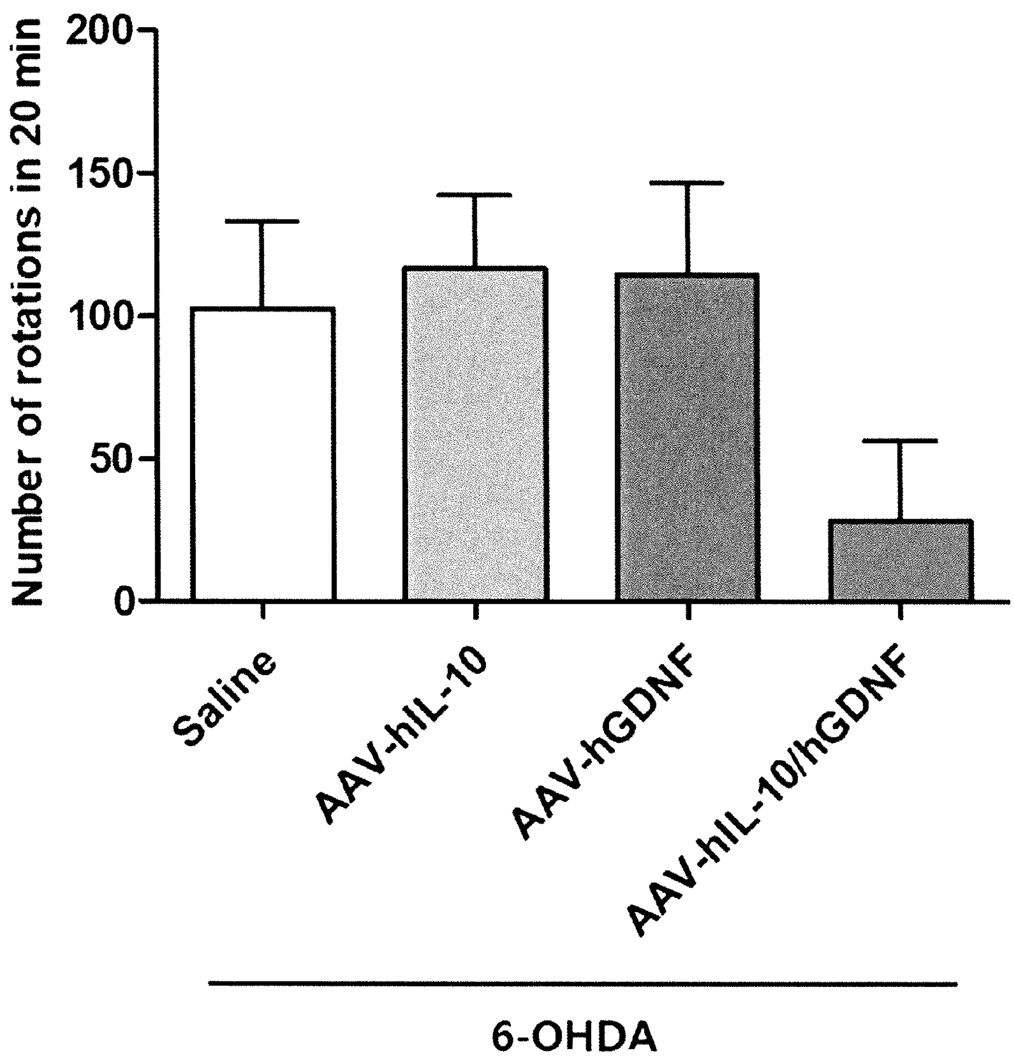
【FIG.1A】


6-OHDA
Saline
AAV-hIL-10
AAV-hGDNF
AAV-hIL-10/hGDNF
Striatum
TH
Scale bar = 500 μm
***
**
**
Striatum optical density
(% of contralateral side)
110
100
90
80
70
0
** p<0.01
***p<0.001
Saline
AAV-hIL-10
AAV-hGDNF
AAV-hIL-10/hGDNF
6-OHDA

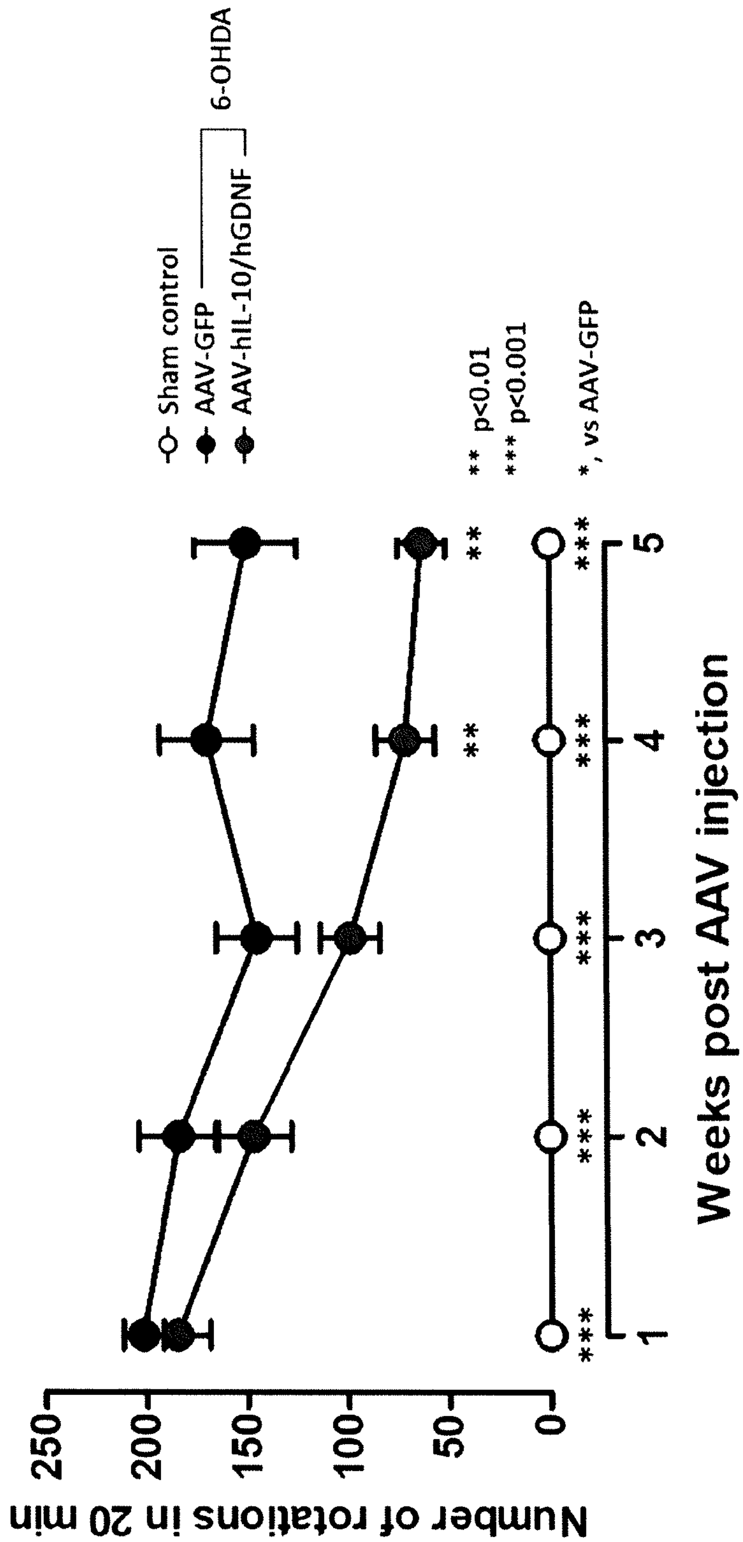
【FIG.1B】


6-OHDA
Saline
AAV-hIL-10
AAV-hGDNF
AAV-hIL-10/hGDNF
TH
SNpc
Scale bar = 500 μm
***
***
***
TH positive cells in SNpc
(% of contralateral side)
150
125
100
60
30
** p<0.01
***p<0.001
Saline
AAV-hIL-10
AAV-hGDNF
AAV-hIL-10/hGDNF
6-OHDA

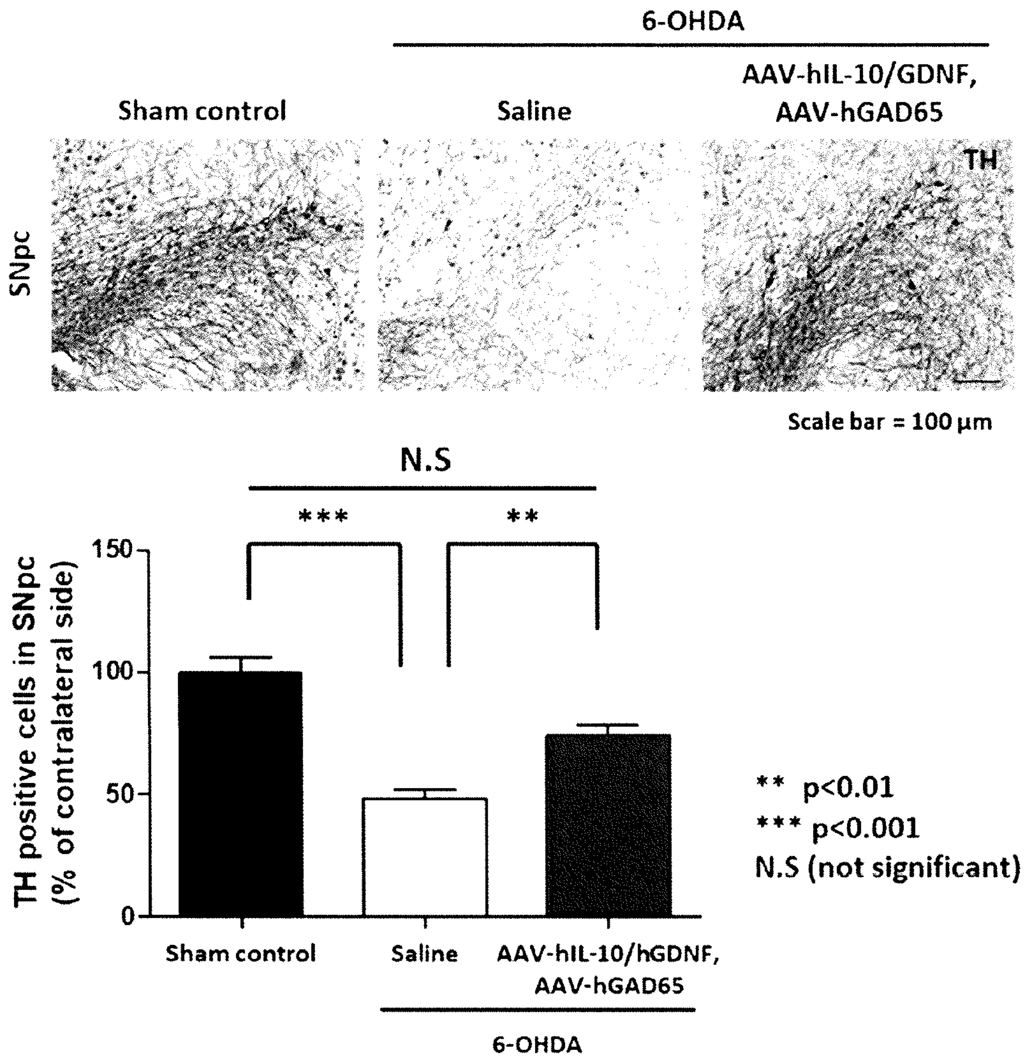
【FIG.1C】


Number of rotations in 20 min
200
150
100
50
0
Saline
AAV-hIL-10
AAV-hGDNF
AAV-hIL-10/hGDNF
6-OHDA

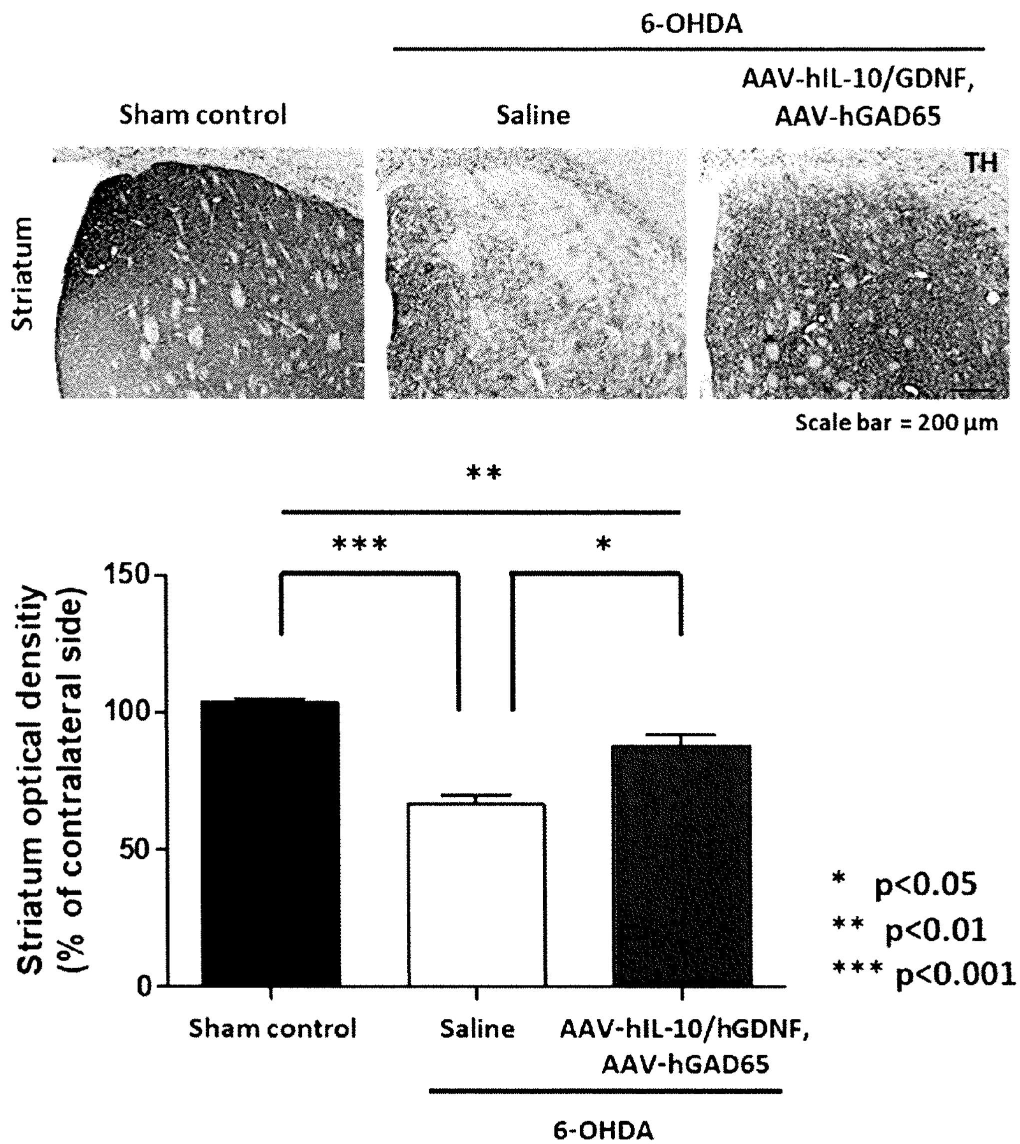
【FIG.2】


Sham control
AAV-GFP
AAV-hIL-10/hGDNF
6-OHDA
** p<0.01
*** p<0.001
*, vs AAV-GFP
Number of rotations in 20 min
0
50
100
150
200
250
1
2
3
4
5
Weeks post AAV injection
***
***
***
***
***
**
**

【FIG.3A】

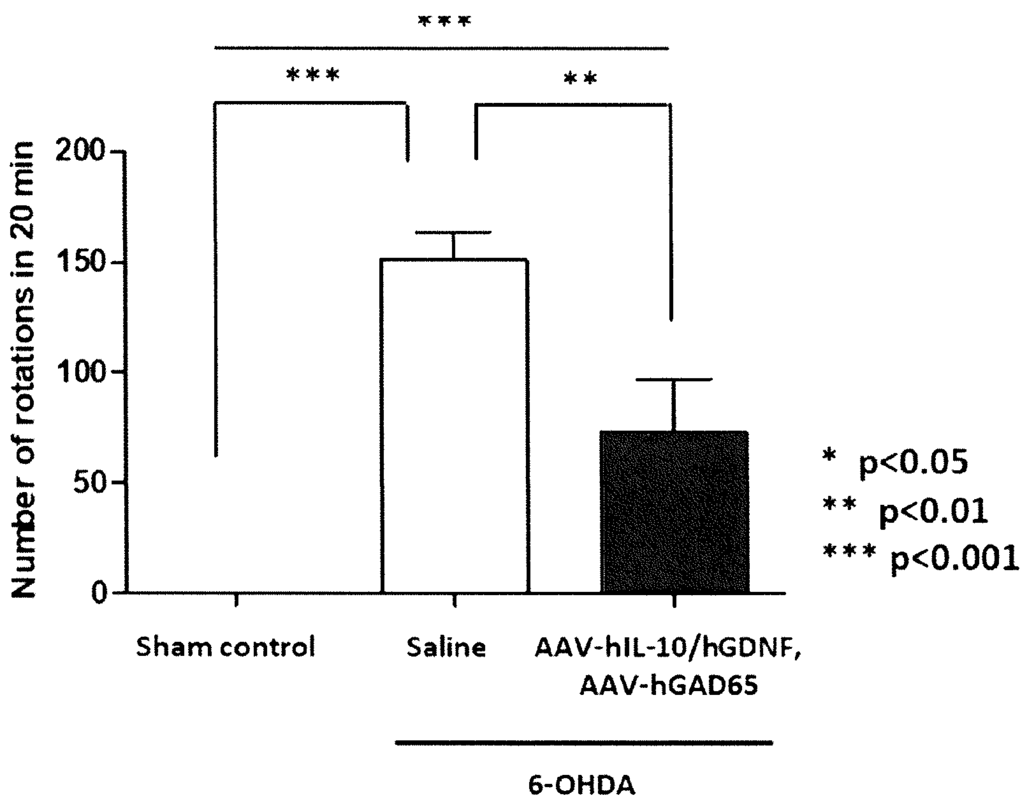

6-OHDA
Sham control
Saline
AAV-hIL-10/GDNF,
AAV-hGAD65
TH
SNpc
Scale bar = 100 μm
N.S
***
**
TH positive cells in SNpc
(% of contralateral side)
150
100
50
0
Sham control
Saline
AAV-hIL-10/hGDNF,
AAV-hGAD65
6-OHDA
** p<0.01
*** p<0.001
N.S (not significant)

【FIG.3B】

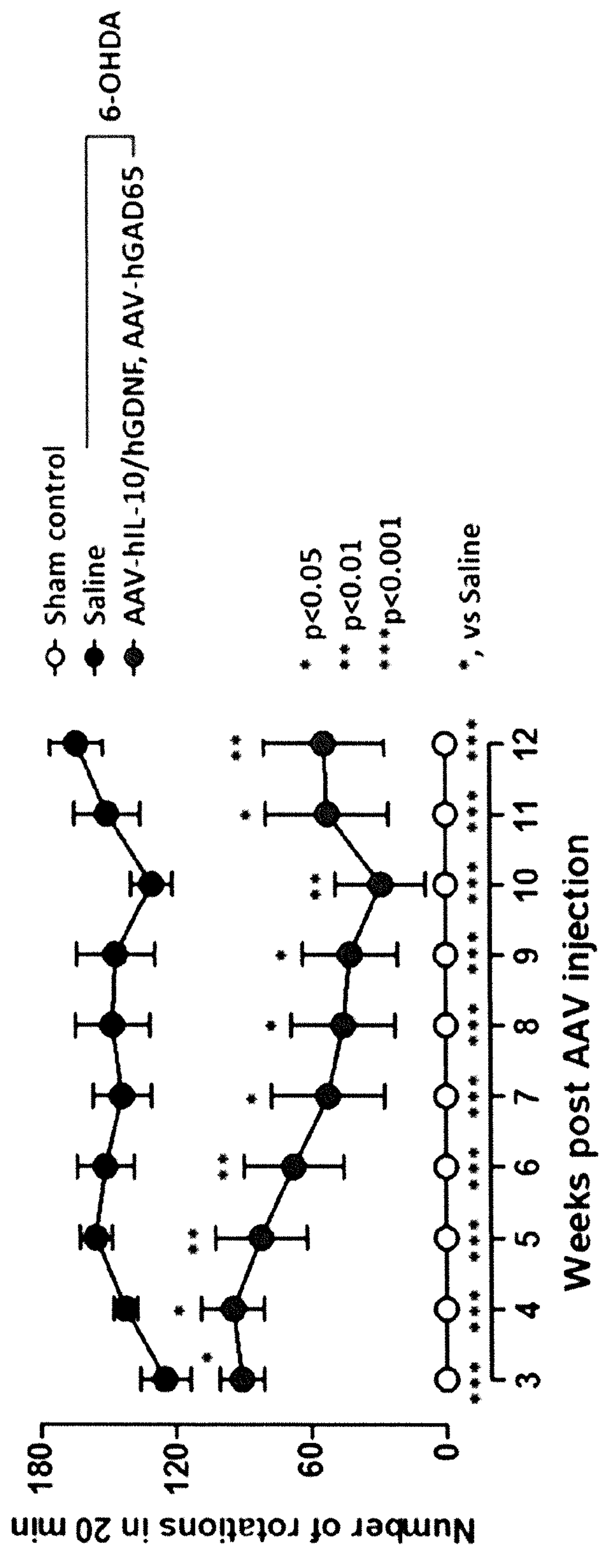


6-OHDA
Sham control
Saline
AAV-hIL-10/GDNF,
AAV-hGAD65
TH
Striatum
Scale bar = 200 μm
**
***
*
150
100
50
0
Striatum optical densitiy
(% of contralateral side)
Sham control
Saline
AAV-hIL-10/hGDNF,
AAV-hGAD65
6-OHDA
* p<0.05
** p<0.01
*** p<0.001

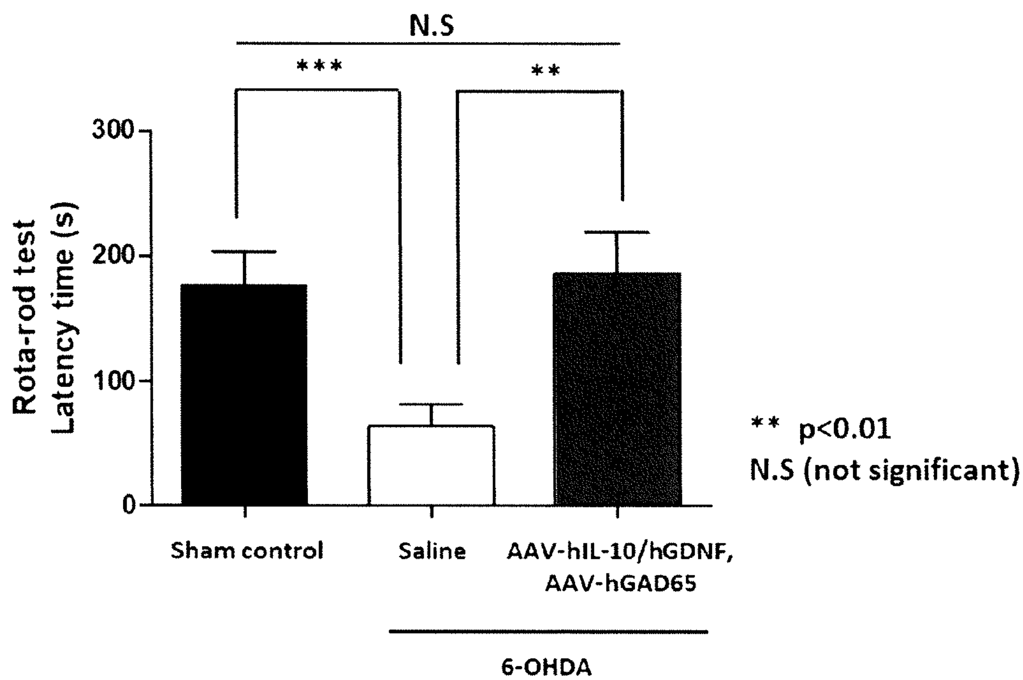
【FIG.4A】


***
***
**
200
150
100
50
0
Number of rotations in 20 min
Sham control
Saline
AAV-hIL-10/hGDNF,
AAV-hGAD65
6-OHDA
* p<0.05
** p<0.01
*** p<0.001

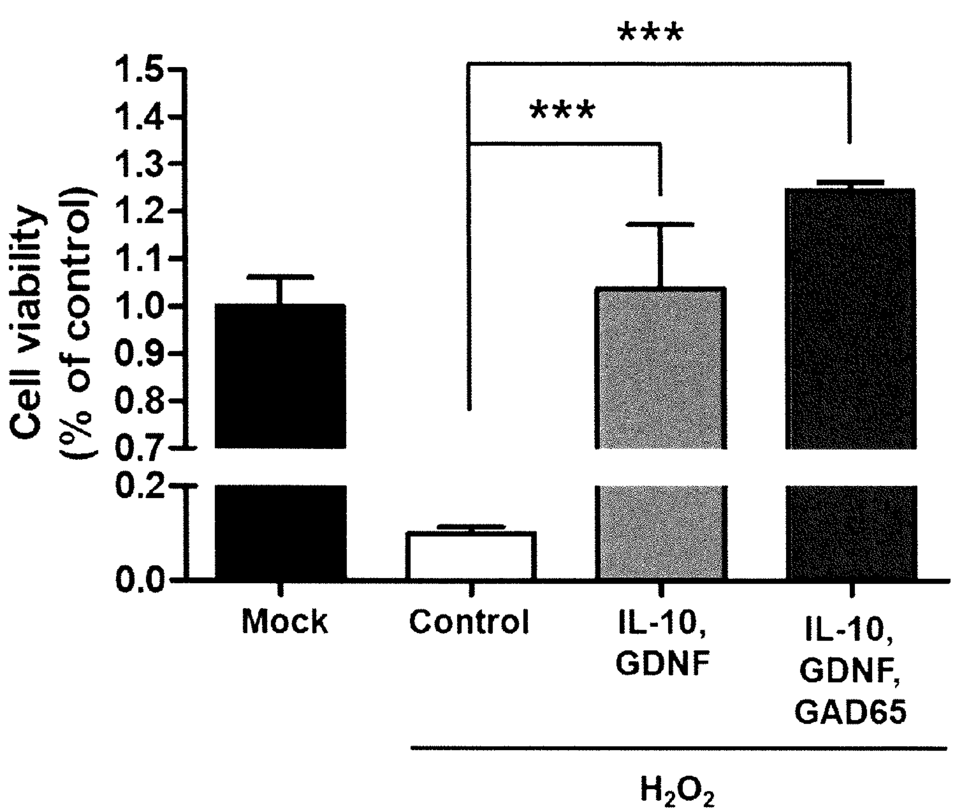
【FIG.4B】


Sham control
Saline
AAV-hIL-10/hGDNF, AAV-hGAD65
6-OHDA
* p<0.05
** p<0.01
*** p<0.001
*, vs Saline
Number of rotations in 20 min
180
120
60
0
3
4
5
6
7
8
9
10
11
12
Weeks post AAV injection

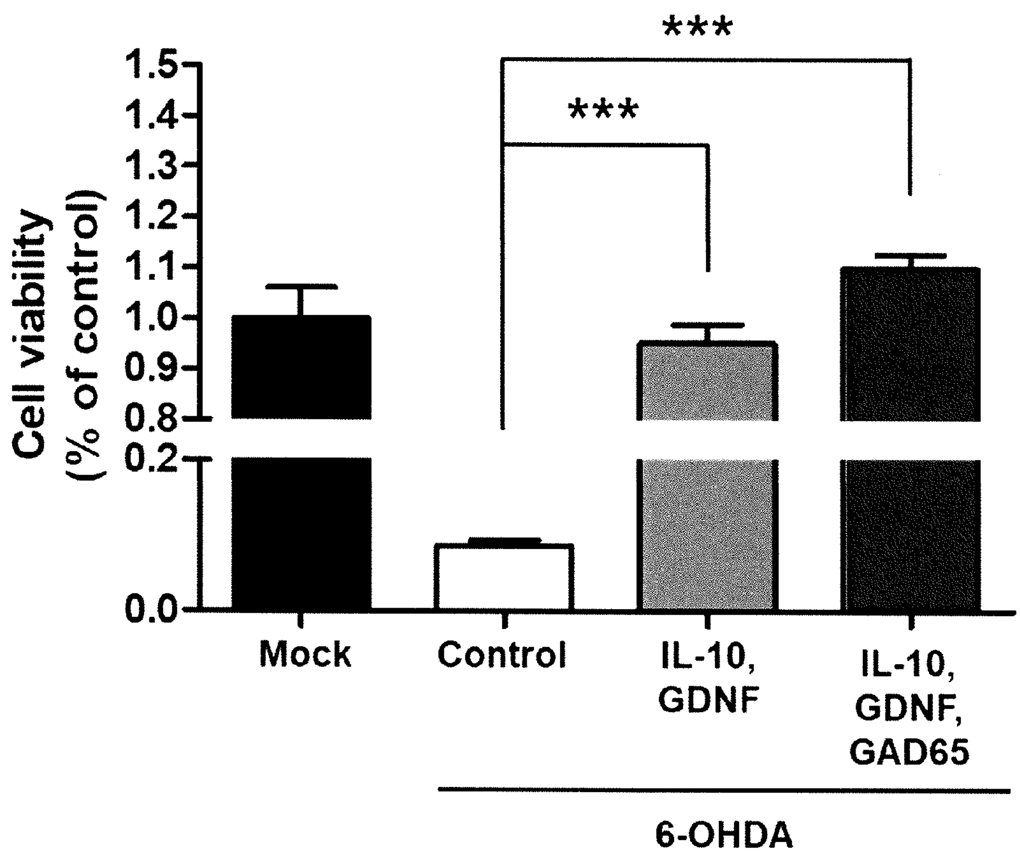
【FIG.4C】


N.S
***
**
300
200
100
0
Rota-rod test
Latency time (s)
** p<0.01
N.S (not significant)
Sham control
Saline
AAV-hIL-10/hGDNF,
AAV-hGAD65
6-OHDA

【FIG.5A】


***
***
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.2
0.0
Cell viability
(% of control)
Mock
Control
IL-10,
GDNF
IL-10,
GDNF,
GAD65
H_2O_2

【FIG.5B】


***
***
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.2
0.0
Cell viability
(% of control)
Mock
Control
IL-10,
GDNF
IL-10,
GDNF,
GAD65
6-OHDA

COMPOSITION FOR PREVENTING OR TREATING DEGENERATIVE BRAIN DISEASES

TECHNICAL FIELD

[0001] The present invention relates to a composition for preventing or treating neurodegenerative diseases and a method for preventing or treating neurodegenerative diseases using the same.

BACKGROUND ART

[0002] A neurodegenerative disease is currently one of the most socially problematic diseases, with a growing elderly population. However, there is no effective therapeutic agent (=drugs) to treat this disease, thus peoples are paying attention to develop prevention and treatment of neurodegenerative disease. A neurodegenerative disease is a general term for diseases caused by the death of neurons, which is progressing slowly and steadily in one or more parts of the nervous system, and a neurodegenerative disease may occur when the brain is damaged by the structural deterioration of neuronal cell in the brain due to aging, secondary symptoms caused by adult diseases such as circulatory disorders, or physical and mechanical factors such as traffic accidents, industrial accidents, and carbon monoxide poisoning, or even when brain tissue is irreversibly damaged by reduced or blocked blood flow to the brain, but the exact cause of disease is currently unknown. When a neurodegenerative disease occurs, progressive cognitive dysfunction, progressive ataxia, muscle weakness, amyotrophy, and the like symptoms may occur depending on the location of the lesion in the brain and pathogenesis of the disease, and the like. Each of various genes is known to be effective in suppressing neuronal cell death. However, single gene therapy has a limitation in that it cannot sufficiently suppress the death of neuronal cells, and since a high dose with single gene therapy is injected to achieve the therapeutic effect, it is not free from side effects such as toxicity. Therefore, there is need for developing a therapy not only effectively preventing or treating neurodegenerative diseases without side effects, but also exhibiting a synergistic effect on the preventive or therapeutic effects of neurodegenerative diseases according to the co-administration of therapeutic genes.

DISCLOSURE

Technical Problem

[0003] Under the background as described above, the present inventors conducted intensive studies to develop a therapy having preventive or therapeutic effects against neurodegenerative diseases, and as a result, confirmed that when at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same are used, the case shows a remarkably excellent preventive or therapeutic effect on neurodegenerative diseases compared to a use of only one, thereby completing the present invention.

[0004] Thus, an object of the present invention is to provide a pharmaceutical composition for preventing or treating a neurodegenerative diseases, containing at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

[0005] Further, another object of the present invention is to provide a health functional food for preventing or ameliorating a neurodegenerative disease, containing at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

[0006] In addition, still another object of the present invention is to provide a method for preventing or treating a neurodegenerative disease, the method including administering at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same to an individual in need thereof.

[0007] Furthermore, yet another object of the present invention is to provide a use of at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same for preparing a preventive or therapeutic pharmaceutical agent for neurodegenerative diseases.

[0008] However, technical problems to be solved by the present invention are not limited to the aforementioned problems, and other problems that are not mentioned may be clearly understood by those skilled in the art from the following description.

Technical Solution

[0009] To achieve the objects, the present invention provides a pharmaceutical composition for preventing or treating a neurodegenerative disease, containing at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

[0010] As an exemplary embodiment of the present invention, the pharmaceutical composition may contain the IL-10 protein or the gene encoding the same and the GDNF protein or the gene encoding the same.

[0011] As another exemplary embodiment of the present invention, the pharmaceutical composition may further contain the GAD protein or the gene encoding the same.

[0012] As still another exemplary embodiment of the present invention, the gene may be contained in a vector.

[0013] As yet another exemplary embodiment of the present invention, the vector may be a viral or non-viral vector.

[0014] As yet another exemplary embodiment of the present invention, the viral vector may be at least one selected from the group consisting of adenoviruses, adeno-associated viruses (AAV), herpes simplex virus, lentiviruses, retroviruses, cytomegalovirus, baculoviruses, and poxviruses.

[0015] As yet another exemplary embodiment of the present invention, the non-viral vector may be at least one selected from the group consisting of plasmids, liposomes, cationic polymers, micelles, emulsions, and solid lipid nanoparticles.

[0016] As yet another exemplary embodiment of the present invention, the IL-10 protein may be an amino acid sequence represented by SEQ ID NO: 1.

[0017] As yet another exemplary embodiment of the present invention, the IL-10 encoding gene may be a base sequence represented by SEQ ID NO: 2 or 3.

[0018] As yet another exemplary embodiment of the present invention, the GDNF protein may be an amino acid sequence represented by SEQ ID NO: 4.

[0019] As yet another exemplary embodiment of the present invention, the GDNF protein encoding gene may be a base sequence represented by SEQ ID NO: 5 or 6.

[0020] As yet another exemplary embodiment of the present invention, the GAD may be GAD 65 or GAD 67.

[0021] As yet another exemplary embodiment of the present invention, the GAD protein may be an amino acid sequence represented by SEQ ID NO: 7 or 9.

[0022] As yet another exemplary embodiment of the present invention, the GAD encoding gene may be a base sequence represented by SEQ ID NO: 8 or 10.

[0023] As yet another exemplary embodiment of the present invention, the neurodegenerative diseases may be at least one selected from the group consisting of Parkinson's disease, dementia, Alzheimer's disease, Huntington's disease, frontotemporal lobar degeneration, dementia with Lewy bodies, vascular dementia, corticobasal degeneration, multiple system atrophy, progressive supranuclear palsy, primary lateral sclerosis, spinal muscular atrophy, stroke, cerebral infarction, head trauma, spinal cord injury, cerebral arteriosclerosis, Lou Gehrig's disease, and multiple sclerosis disease.

[0024] As yet another exemplary embodiment of the present invention, the pharmaceutical composition may additionally contain a physiologically acceptable carrier.

[0025] As yet another exemplary embodiment of the present invention, the pharmaceutical composition may be an injection.

[0026] As yet another exemplary embodiment of the present invention, the pharmaceutical composition may have an effect of ameliorating motor abnormalities caused by neuronal cell death.

[0027] As yet another exemplary embodiment of the present invention, the neurons may be motor neurons, interneurons or sensory neurons.

[0028] As yet another exemplary embodiment of the present invention, the pharmaceutical composition may have an effect of improving cognitive ability and memory.

[0029] Further, the present invention provides a health functional food for preventing or ameliorating neurodegenerative diseases, containing at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

[0030] In addition, the present invention provides a method for preventing or treating neurodegenerative diseases, the method including administering at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same to an individual in need thereof.

[0031] Furthermore, the present invention provides a use of at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same for preparing a preventive or therapeutic pharmaceutical agent for neurodegenerative diseases.

Advantageous Effects

[0032] The composition of the present invention contains at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same. Therefore, since the pharmaceutical composition of the present invention exhibits a better effect of alleviating and preventing neurodegenerative diseases by co-administration of active ingredients than upon single administration, the pharmaceutical composition of the present invention can exhibit the same or higher effect only with a low concentration of the active ingredients. Thus, the pharmaceutical composition of the present invention can be advantageously used for the prevention or treatment of neurodegenerative diseases.

[0033] However, the effects of the present invention are not limited to the aforementioned effects and should be understood to include all possible effects deduced from the configuration of the invention described in the detailed description or the claims of the present invention.

DESCRIPTION OF DRAWINGS

[0034] FIG. **1** is a result confirming the effect of co-administration of hIL-10 and hGDNF on ameliorating neurodegenerative diseases, FIGS. **1**A and **1**B are results confirming that, in a 6-OHDA induced neurodegenerative animal models pretreated with test articles, rAAV5-hIL-10/hGDNF administration (co-administration) is more effective in protecting dopaminergic nerve fiber than the administration of rAAV5-hIL-10 and rAAV5-hGDNF alone in the striatum region of the brain, FIG. **1**B is a result confirming that, in a 6-OHDA induced neurodegenerative animal models pretreated with test articles, rAAV5-hIL-10/hGDNF administration (co-administration) is more effective in protecting dopaminergic neurons than the administration of rAAV5-hIL-10 and rAAV5-hGDNF alone in the substantia nigra pars compacta (SNpc) region, and FIG. **1**C is a result confirming, through a rotation test, that rAAV5-hIL-10/hGDNF administration (co-administration) is more effective in ameliorating hypokinesia symptoms than the administration of rAAV5-hIL-10 and rAAV5-hGDNF alone.

[0035] FIG. **2** is a result confirming the long-term effect of ameliorating motor function in a neurodegenerative disease animal model. rAAV5-hIL-10/hGDNF was injected into 6-OHDA induced neurodegenerative disease model and its efficacy in improving motor function was confirmed using a rotation test.

[0036] FIG. **3** is a result confirming the neuroprotective effect of the gene therapy of the present invention on dopaminergic neurons and fibers. One of the gene therapies of the present invention (rAAV5-hIL-10/hGDNF and rAAV5-hGAD65) was pretreated and neurodegenerative disease animal model was induced by 6-OHDA to confirm the efficacy of dopaminergic neuroprotection. Specifically, FIG. **3**A shows the result confirmed in the striatum, and FIG. **3**B shows the result confirmed in the substantia nigra pars compacta (SNpc).

[0037] FIG. **4** showed the hypokinesia prevention effect of the gene therapy of the present invention in 6-OHDA induced neurodegenerative mouse model by pretreating one of the present invention (rAAV5-hIL-10/hGDNF and rAAV5-hGAD65). Specifically, FIGS. **4***a* and **4***b* is results of rotation test after inducing animal disease model at 4 and 10 weeks, respectively. FIG. **4***c* showed the results of Rota-rod test, 2 weeks post inducing animal disease model.

[0038] FIG. **5** is a set of results confirming the anti-apoptotic effect on the oxidative stress in co-administration of hIL-10 and hGDNF supernatant or simultaneous administration of hIL-10, hGDNF, and hGAD65 supernatant. The combination supernatant was produced by each therapeutic gene plasmid transfection. Specifically, FIG. **5**A and FIG. **5**B are confirming that the effect of inhibiting neuronal cell death is exhibited in co-administration of hIL-10 and hGDNF supernatant or simultaneous administration of hIL-10, hGDNF, and hGAD65 supernatant in the case of neuronal cell death by H_2O_2 and 6-OHDA, respectively.

MODES OF THE INVENTION

[0039] The present inventors conducted intensive studies to develop a therapeutic pharmaceutical t having preventive or therapeutic effects against neurodegenerative diseases, and as a result, confirmed that when at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same are used, the case shows an excellent preventive or therapeutic effect on neurodegenerative diseases compared to a use of only one, thereby completing the present invention.

[0040] Thus, the present invention provides a pharmaceutical composition for preventing or treating neurodegenerative diseases, containing at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

[0041] As used herein, the term "prevention" refers to all actions that suppress or delay the onset of a neurodegenerative disease by administering the pharmaceutical composition provided by the present invention to an individual expected to develop the neurodegenerative disease.

[0042] As used herein, the term "treatment" refers to all actions that clinically intervene to alter the natural processes of the individual or cells to be treated, and may be carried out either during the development of a clinical pathological condition or to prevent the clinical pathological condition. The desired therapeutic effect includes preventing the occurrence or recurrence of a disease, alleviating symptoms, reducing all direct or indirect pathological consequences of the disease, preventing metastasis, reducing a disease progression rate, mitigating or temporarily alleviating a condition, making a patient feel better, or improving the prognosis. For the purpose of the present invention, the treatment may be interpreted as including all actions of administering the pharmaceutical composition of the present invention to a patient who has developed a neurodegenerative disease to ameliorate the symptoms of the neurodegenerative disease, but is not limited thereto.

[0043] A combination of the two or more may be interleukin 10 (IL-10) and glial cell line-derived neurotrophic factor (GDNF); interleukin 10 (IL-10) and glutamate decarboxylase (GAD); glial cell line-derived neurotrophic factor (GDNF) and glutamate decarboxylase (GAD); interleukin 10 (IL-10), glial cell line-derived neurotrophic factor (GDNF) and glutamate decarboxylase (GAD), and the IL-10, GDNF and GAD may be each a protein or gene encoding the same.

[0044] Further, the gene may be DNA or RNA (specifically mRNA), and may be in the form of being operably contained in a vector. Specifically, the gene may be in the form of being operably included in a viral or a non-viral vector.

[0045] The gene encoding IL-10 may be operably included in a first vector, the gene encoding GDNF may be operably included in a second vector, the gene encoding GAD may be operably included in a third vector, and two or more genes may be included in one vector.

[0046] The two or more genes selected from the group consisting of the gene encoding IL-10, the gene encoding GDNF, and the gene encoding GAD may be in the form of being included in a vector. In this case, the vector may be a viral vector, or a non-viral vector such as a plasmid or a liposome. In addition, some of the genes may be in the form of being included in a viral vector and the remaining genes may be included in the form of being included in a non-viral vector.

[0047] As an exemplary embodiment, IL-10 may be in the form of being included in a viral vector, and GDNF may be in the form of being included in a non-viral vector. Furthermore, GDNF may be in the form of being included in a viral vector, and IL-10 may be in the form of being included in a non-viral vector. Moreover, IL-10 may be in the form of being included in a viral vector, and GAD may be in the form of being included in a non-viral vector. Further, IL-10 may be in the form of being included in a viral vector, and GDNF and GAD may be in the form of being included in a non-viral vector. In addition, IL-10 and GDNF may be in the form of being included in a viral vector, and GAD may be in the form of being included in a non-viral vector. Furthermore, IL-10 and GAD may be in the form of being included in a viral vector, and GDNF may be in the form of being included in a non-viral vector. Further, IL-10 may be in the form of being included in a viral vector, and GDNF and GAD may be in the form of being included in a non-viral vector. In addition, GDNF and GAD may be in the form of being included in a viral vector, and IL-10 may be in the form of being included in a non-viral vector. Furthermore, GDNF may be in the form of being included in a viral vector, and IL-10 and GAD may be in the form of being included in a non-viral vector. Further, all of IL-10, GDNF and GAD may be in the form of being included in a viral vector. All of IL-10, GDNF and GAD may be in the form of being included in a non-viral vector.

[0048] The viral vector may be at least one selected from the group consisting of adenoviruses, adeno-associated viruses (AAV), herpes simplex virus, lentiviruses, retroviruses, cytomegalovirus, baculoviruses, poxviruses, and the like, but is not limited thereto. Preferably, the viral vector may be an adeno-associated virus.

[0049] In addition, the non-viral vector may be at least one selected from the group consisting of plasmids, liposomes, cationic polymers, micelles, emulsions, and solid lipid nanoparticles.

[0050] As used herein, the term "plasmid" refers to a circular DNA fragment that is present separately from the bacterial chromosome. The plasmid does not include genes that are essential for bacterial survival, but it includes genes that are essential for resistance to certain antibiotics and gene exchange between bacteria. Furthermore, the plasmid can proliferate independently of chromosomes, and may include a selectable marker.

[0051] As used herein, the term "liposome" refers to a vesicle formed while forming a bilayer due to a hydrophilic part and a hydrophobic part when molecules having both the hydrophobic part and the hydrophilic part, such as phospholipids, are suspended in an aqueous solution. Liposomes are separated from the outer membrane by a membrane composed of a lipid bilayer, and liposomes incorporating DNA, mRNA, and the like may be used as carriers of genetic information.

[0052] As used herein, the term "cationic polymer" refers to a material that forms a complex with anionic DNA through ionic bonding as a cationic lipid or polymer compound, and is delivered into cells.

[0053] As used herein, the term "micelle" refers to a thermodynamically stable colloidal aggregate made by the association of molecules consisting of polar groups and non-polar groups, such as surfactants and lipid molecules, by van der Waals forces in a solution. Further, micelles containing DNA, RNA, and the like may be used as carriers of genetic information.

[0054] As used herein, the term "emulsion" means that, when two solutions having different phases are mixed, one liquid forms fine particles and is dispersed in the other liquid. DNA, mRNA and the like may be included in the core of the emulsion particles and used as carriers of genetic information.

[0055] As used herein, the term "solid lipid nanoparticles" refers to a formulation in which a drug is incorporated in nano-sized microparticles consisting of a solid lipid instead of a liquid lipid.

[0056] A mixing ratio of Vector 1 (for example, a first vector) including any one gene selected from the group consisting of IL-10, GDNF and GAD and Vector 2 (for example, a second vector) including at least one gene selected from the group consisting of the remaining genes, which are not included in Vector 1, may be 1:1 to 100 or 1 to 100:1. Specifically, although not limited thereto, the mixing ratio of Vector 1 and Vector 2 may be 1:1 to 100 or 1 to 100:1 based on VG.

[0057] Vector 1 (for example, a first vector) including the gene encoding IL-10, Vector 2 (for example, a second vector) including the gene encoding GDNF, and Vector 3 (for example, a third vector) including the gene encoding GAD may be mixed in various ratios, and for example, the mixing ratio may be 1:1:1 to 1:1:100; 1:1:1 to 1:100:1; 1:1:1 to 100:1:1; 1:1:1 to 1:100:100; 1:1:1 to 100:100:1; or 1:1:1 to 100:1:100, but is not limited thereto.

[0058] As used herein, the term "operably" refers to being linked to a regulatory sequence in a manner that allows a transgene to be expressed in a host cell. The regulatory sequence is a DNA sequence that regulates gene expression, and may include promoters and other regulatory elements such as enhancers and polyadenylation. In addition, the regulatory sequence provides a binding site for the transcription factor that regulates the expression of the transgene, and may affect a complex structure with the transcription factor to determine the function of the transcription factor.

[0059] As used herein, the term "IL-10" refers to an anti-inflammatory cytokine belonging to class II cytokines (Renauld, Nat Rev Immunol, 2003). IL-10 is in the form of a homodimer consisting of two subunits, each with a length of 178 amino acids, and is also known as a human cytokine synthesis inhibitory factor (CSIF). IL-10 performs the function of suppressing the activity of natural killer (NK) cells in immune responses and forms a complex with the IL-10 receptor to participate in signal transduction. IL-10 may be a human-, mouse-, rat-, dog-, cat-, or horse-derived protein or a base sequence encoding the same, but is not limited thereto. IL-10 may consist of an amino acid sequence of SEQ ID NO: 1 of NCBI Accession No. NP_000563.1, and the gene encoding the same may be a base sequence represented by SEQ ID NO: 2 or SEQ ID NO: 3.

[0060] The IL-10 encoding gene may be a base sequence encoding an IL-10 variant that maintains IL-10 activity. The base sequence encoding the IL-10 variant may be a base sequence encoding an amino acid sequence having a sequence homology of at least 60% or more, 70% or more, 80% or more, or 90% or more with the IL-10 amino acid sequence indicated above, and may be most preferably a base sequence encoding an amino acid sequence having a sequence homology of 95% or more.

[0061] Furthermore, the base sequence encoding the IL-10 variant may be a base sequence having a sequence homology of at least 60% or more, 70% or more, 80% or more, or 90% or more with the IL-10 base sequence indicated above, and may be most preferably a base sequence having a sequence homology of 95% or more.

[0062] The "% sequence homology" is confirmed by comparing comparison regions with two sequences optimally aligned, and a portion of the base sequence in the comparison region may include an addition or deletion (that is, a gap) compared to the reference sequence (without addition or deletion) for the optimal alignment of the two sequences.

[0063] As used herein, the term "GDNF" refers to one of nerve growth factor that constitutes the GDNF ligand family, and the GDNF ligand family includes GDNF, neuturin (NRTN), artemin (ARTN), and persephin (PSPN). Further, GDNF is a protein that promotes the survival of many types of neurons, and transmits signals through GDNF receptor alpha-1 (GFR α1), GDNF receptor alpha-2 (GFR α2) and GDNF receptor alpha-3 (GFR α3). GDNF may be a human-, mouse-, rat-, dog-, cat-, or horse-derived protein or a base sequence encoding the same, but is not limited thereto. GDNF may consist of an amino acid sequence of SEQ ID NO: 4 of NCBI Accession No. NCBI NP_954701.1, and the gene encoding the same may be a base sequence represented by SEQ ID NO: 5 or SEQ ID NO: 6.

[0064] The GDNF encoding gene may be a base sequence encoding a GDNF variant that maintains GDNF activity. The base sequence encoding the GDNF variant may be a base sequence encoding an amino acid sequence having a sequence homology of at least 60% or more, 70% or more, 80% or more, or 90% or more with the GDNF amino acid sequence indicated above, and may be most preferably a base sequence encoding an amino acid sequence having a sequence homology of 95% or more.

[0065] Further, the base sequence encoding the GDNF variant may be a base sequence having a sequence homology of at least 60% or more, 70% or more, 80% or more, or 90% or more with the GDNF base sequence indicated above, and may be most preferably a base sequence having a sequence homology of 95% or more.

[0066] As used herein, the term "GAD" refers to an enzyme that decarboxylates glutamic acid to produce gamma-aminobutyric acid (GABA). The GAD may be GAD65 or GAD67. Specifically, the GAD65 may be a human-, mouse-, rat-, dog-, cat-, or horse-derived protein or a base sequence encoding the same, but is not limited thereto. The GAD65 may consist of an amino acid sequence of SEQ ID NO: 7 of NCBI Accession No. NP_000809.1, and the gene encoding the same may be a base sequence represented by SEQ ID NO: 8. The GAD67 may be a human-, mouse-, rat-, dog-, cat-, or horse-derived protein or a base sequence encoding the same, but is not limited thereto. The GAD67 may consist of an amino acid sequence of SEQ ID NO: 9 of NCBI Accession No. NP_000808.2, and the gene encoding the same may be a base sequence represented by SEQ ID NO: 10.

[0067] In addition, the GAD encoding gene may be a base sequence that encodes a GAD variant that enables GABA to be produced by maintaining GAD activity. The GAD variant includes all sequences in which the characteristics of GAD producing GABA are maintained, and is not limited to thereto, but preferably, the base sequence encoding the GAD variant may be a base sequence encoding an amino acid sequence having a sequence homology of at least 60% or more, 70% or more, 80% or more, or 90% or more with the GAD amino acid sequence indicated above, and may be most preferably a base sequence encoding an amino acid sequence having a sequence homology of 95% or more.

[0068] In addition, the base sequence encoding the GAD variant may be a base sequence having a sequence homology of at least 60% or more, 70% or more, 80% or more, or 90% or more with the GAD base sequence indicated above, and may be most preferably a base sequence having a sequence homology of 95% or more.

[0069] The pharmaceutical composition of the present invention may exhibit a preventive effect against neurodegenerative diseases through a therapeutic genes or vectors containing the genes. The composition of the present invention may achieve a better neurodegenerative disease preventive or therapeutic effect than single administration by co-administering therapeutic pharmaceutical agents having different mechanisms.

[0070] In particular, according to the present invention, when two or more genes selected from the group consisting of genes encoding IL-10, GDNF and GAD65 are co-administered, a synergistic preventive or therapeutic effect on neurodegenerative diseases shown. Therefore, the pharmaceutical composition of the present invention may be advantageously used for the prevention or treatment of neurodegenerative diseases, and specifically may ameliorate motor abnormalities in neurodegenerative diseases caused by the death of neurons, and may ameliorate cognitive and memory decline, which are the main symptoms of neurodegenerative diseases.

[0071] In the present invention, each of the vectors including IL-10, GDNF and GAD65 may be administered alone or two or more thereof may be administered together, and may also be administered in a form including two or more genes in one vector. As an administration site, IL-10 may be administered to the striatum (ST), GDNF may be administered to the striatum (ST), and GAD65 may be administered to the subthalamic nucleus (STN), but the administration is not limited thereto. The administration site of rAAV5-hGDNF/hIL-10, which is one of the aforementioned examples, may be the striatum (ST), but is not limited thereto.

[0072] In the present invention, the neurons may be motor neurons that transmit signal to the brain or spinal cord and transmit the signal to effectors such as muscle cells or secretory glands, or interneurons that are distributed throughout the central nervous system and transmit signal from sensory neurons to motor neurons, or sensory neurons that transmit signal to the brain or spinal cord to transmit the signal to effectors such as muscle cells or secretory glands.

[0073] In the present invention, the neurodegenerative disease is not limited to, but may be at least one selected from the group consisting of Parkinson's disease, dementia, Alzheimer's disease, Huntington's disease, frontotemporal lobar degeneration, dementia with Lewy bodies, vascular dementia, corticobasal degeneration, multiple system atrophy, progressive supranuclear palsy, primary lateral sclerosis, spinal muscular atrophy, stroke, cerebral infarction, head trauma, spinal cord injury, cerebral arteriosclerosis, Lou Gehrig's disease, and multiple sclerosis disease, and may be preferably Parkinson's disease, dementia or Alzheimer's disease.

[0074] The present inventors confirmed, through specific exemplary embodiments, that the pharmaceutical composition of the present invention can be advantageously used for the prevention or treatment of neurodegenerative diseases.

[0075] In an exemplary embodiment of the present invention, it was confirmed that, when rAAV5-hIL-10/hGDNF was administered to a mouse model in which a neurodegenerative disease was induced, motor dysfunction, which is a symptom induced by the neurodegenerative disease, was statistically significant ameliorated (see Example 1).

[0076] In another exemplary embodiment of the present invention, it is confirmed that pre-treatment of rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 on 6-OHDA induced neurodegenerative disease model has been found to ameliorate dopaminergic neuronal cell death and motor dysfunction, which are symptoms of neurodegenerative disease. Improvement of these neurodegenerative symptoms is induced by administration of the pharmaceutical composition of the present invention. (see Example 2).

[0077] In still another embodiment of the present invention, it was confirmed that in the case of pretreatment with hIL-10 and hGDNF supernatants produced by each gene plasmid or hIL-10, hGDNF and hGAD65 supernatants produced by each gene plasmid, neuronal cell death induced by H_2O_2 or 6-OHDA was suppressed (see Example 3).

[0078] Through the aforementioned results, the present inventors were able to confirm that neurodegenerative diseases can be prevented or treated by the pharmaceutical composition of the present invention.

[0079] The pharmaceutical composition of the present invention may further include a suitable carrier, excipient, and diluent, which are typically used to prepare a pharmaceutical composition. A composition of pharmaceutically acceptable carrier can be in various forms, such as oral or parenteral formation. When the composition is formulated, the composition may be produced using a commonly used diluent or excipient, such as a filler, an extender, a binder, a wetting agent, a disintegrant, and a surfactant. A solid formulation for oral administration may include a tablet, a pill, a powder, a granule, a capsule, and the like, and the solid formulation may be prepared by mixing at least one excipient, for example, starch, calcium carbonate, sucrose or lactose, gelatin, and the like with one or more compounds.

[0080] Furthermore, in addition to simple excipients, lubricants such as magnesium stearate and talc may also be used. A liquid formulation for oral administration corresponds to a suspension, an emulsion, a syrup, and the like, and the liquid formulation may include, in addition to water and liquid paraffin which are simple commonly used diluents, various excipients, for example, a wetting agent, a sweetener, an aromatic, a preservative, and the like. A formulation for parenteral administration may include an aqueous sterile solution, a non-aqueous solvent, a suspension, an emulsion, a freeze-dried preparation, and a suppository. As the non-aqueous solvent and the suspension solvent, it is possible to use propylene glycol, polyethylene glycol, a vegetable oil such as olive oil, an injectable ester such as ethyl oleate, and the like. As a base of the suppository, it is possible to use Witepsol, Macrogol, Tween 61, cacao butter, laurin fat, glycerogelatin, and the like.

[0081] Further, the pharmaceutical composition of the present invention may have any one formulation selected from the group consisting of a tablet, a pill, a powder, a granule, a capsule, a suspension, an emulsion, a syrup, a sterilized aqueous solution, a non-aqueous solvent, a suspension, an emulsion, a freeze-dried preparation, and a suppository.

[0082] Meanwhile, as another aspect of the present invention, the present invention provides a method for preventing or treating neurodegenerative diseases, the method including administering the pharmaceutical composition to an individual.

[0083] As used herein, the term “individual” refers to all animals, including humans, who are likely to develop or have developed the neurodegenerative disease. By administering the compositions of the present invention to an individual, symptoms induced by a neurodegenerative disease may be alleviated or treated.

[0084] As used herein, the term “alleviation” refers to all actions that ameliorate or beneficially change symptoms of a neurodegenerative disease by administering the pharmaceutical composition according to the present invention.

[0085] The composition of the present invention is administered in a pharmaceutically effective dose.

[0086] As used herein, the term “administration” refers to introduction of the pharmaceutical composition of the present invention into an individual by any appropriate method, and the pharmaceutical composition of the present invention can be administrated through various routes, either oral or parenteral, that can reach the target tissues.

[0087] The pharmaceutical composition of the present invention may be appropriately administered to an individual according to the typical method, administration route and dosage used in the art, according to the purpose or need. Examples of the administration route include oral, parenteral, subcutaneous, intraperitoneal, intrapulmonary, and intranasal administration, and for parenteral injection includes intramuscular, intravenous, intraarterial, intraperitoneal, or subcutaneous administration.

[0088] In addition, an appropriate dosage and frequency of administration may be selected according to methods known in the art, and the dose and administration frequency of the pharmaceutical composition of the present invention actually administered can be appropriately determined by various factors such as the type of symptoms to be treated, administration route, sex, health status, diet, age and weight of the individual and severity of disease.

[0089] In the case of a viral vector, the administration frequency may be once or more per year, and when the viral vector is repeatedly administered, it may be administered at intervals of 1 day to 1 month, or 1 month to 1 year. Furthermore, when the pharmaceutical composition is a non-viral vector, it may be administered once or more a year, and when the non-viral vector is repeatedly administered, it may be administered at intervals of 12 to 24 hours and 1 to 14 days. Further, when the pharmaceutical composition is a protein, it may be administered once or more a year, and when it is repeatedly administered, it may be administered at intervals of 1 day to 1 month, or 1 month to 1 year.

[0090] As used herein, the term “pharmaceutically effective dose” refers to an dosage sufficient to treat a neurodegenerative disease at a reasonable benefit/risk ratio applicable to medical use, and an effective dosage level may be determined according to factors including the type of individual, the severity of disease, age, sex, the activity of drugs, sensitivity to drugs, administration time, administration route, excretion rate, treatment period, and simultaneously used drugs, and other factors well known in the medical field.

[0091] The pharmaceutical composition of the present invention may be administered as an individual therapeutic pharmaceutical agent or may be administered in combination with other therapeutic pharmaceutical agents, and may be administered sequentially or simultaneously with therapeutic pharmaceutical agents in the related art. Also, the composition of the present invention may be administered in a single dose or multiple doses. It is important to administer the composition in a minimum effective dose that can obtain the maximum effect without any side effects, in consideration of all the aforementioned factors, and this dose may be easily determined by those skilled in the art.

[0092] Further, the present invention provides a health functional food for preventing or ameliorating neurodegenerative diseases, containing at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

[0093] The “interleukin 10,” “glial cell line-derived neurotrophic factor,” “glutamate decarboxylase,” “protein,” “gene,” “neurodegenerative disease,” “prevention,” and the like may fall within the above-described ranges.

[0094] The term “amelioration” may mean all actions that at least reduce a parameter associated with the condition to be treated, for example, the severity of symptoms. In this case, in order to prevent or alleviate a neurodegenerative disease, the health functional food may be used simultaneously with or separately from a drug for treatment before or after the onset of the corresponding disease.

[0095] The health functional food may be used by adding an active ingredient as it is to food or may be used together with other foods or food ingredients, and may be appropriately used according to a typical method. The mixing amount of the active ingredient may be suitably determined depending on its purpose of use (for prevention or alleviation). In general, the health functional food may be added in an amount of about 15% by weight or less, more specifically about 10% by weight or less, based on the raw materials when manufacturing food or beverages. However, for long-term intake for the purpose of health and hygiene or for the purpose of health control, the amount may be below the above-mentioned range.

[0096] The health functional food further includes one or more of a carrier, a diluent, an excipient, and an additive, and thus, may be formulated into one selected from the group consisting of a tablet, a pill, a powder, a granule, a powder, a capsule, and a liquid formulation. Foods to which the compound according to an aspect can be added include various foods, powders, granules, tablets, capsules, syrups, beverages, gums, teas, vitamin complexes, health functional foods, and the like.

[0097] Specific examples of the carrier, excipient, diluent, and additive may be one or more selected from the group consisting of lactose, dextrose, sucrose, sorbitol, mannitol, erythritol, starch, acacia rubber, calcium phosphate, alginate, gelatin, calcium phosphate, calcium silicate, microcrystalline cellulose, polyvinylpyrrolidone, cellulose, polyvinylpyrrolidone, methyl cellulose, water, sugar syrup, methyl cellulose, methyl hydroxybenzoate, propyl hydroxybenzoate, talc, magnesium stearate, and mineral oil.

[0098] In addition to containing the active ingredient, the health functional food may contain other ingredients as essential ingredients without any particular limitation. For example, the health functional food may contain various flavoring agents or natural carbohydrates, and the like as an additional ingredient, as in a typical beverage. Examples of the above-described natural carbohydrates include typical sugars such as monosaccharides, for example, glucose, fructose and the like; disaccharides, for example, maltose, sucrose and the like; and polysaccharides, for example, dextrin, cyclodextrin and the like, and sugar alcohols such as xylitol, sorbitol, and erythritol. As the flavoring agent other than those mentioned above, a natural flavoring agent (thaumatin, a stevia extract (for example, rebaudioside A, glycyrrhizin and the like)), and a synthetic flavoring agent (saccharin, aspartame and the like) may be advantageously used. The proportion of the natural carbohydrate may be appropriately determined by the choice of a person skilled in the art.

[0099] The health functional food according to an aspect may contain various nutrients, vitamins, minerals (electrolytes), flavoring agents such as synthetic flavoring agents and natural flavoring agents, colorants and fillers (cheese, chocolate, and the like), pectic acid and salts thereof, alginic acid and salts thereof, organic acids, protective colloid thickeners, pH adjusting agents, stabilizers, preservatives, glycerin, alcohols, carbonating agents used in carbonated beverages, or the like, in addition to the additives mentioned above. Such components may be used independently or in combination, and the proportion of these additives may also be appropriately selected by those skilled in the art.

[0100] In addition, the present invention provides a method for preventing or treating neurodegenerative diseases, the method including administering at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same to an individual in need thereof.

[0101] The “interleukin 10,” “glial cell line-derived neurotrophic factor,” “glutamate decarboxylase,” “protein,” “gene,” “neurodegenerative disease,” “prevention,” “treatment,” “individual” and the like may fall within the above-described ranges.

[0102] Furthermore, the present invention provides a preparation of at least two selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same for preparing a preventive or therapeutic pharmaceutical agent for neurodegenerative diseases.

[0103] The “interleukin 10,” “glial cell line-derived neurotrophic factor,” “glutamate decarboxylase,” “protein,” “gene,” “neurodegenerative disease,” “prevention,” “treatment” and the like may fall within the above-described ranges.

[0104] Hereinafter, preferred examples for helping with understanding of the present invention will be suggested. However, the following examples are provided only so that the present invention may be more easily understood, and the content of the present invention is not limited by the following examples.

EXAMPLES

Example 1. Confirmation of Synergistic Effect of Co-Administration of hIL-10 and hGDNF on Amelioration of Neurodegenerative Disease

[0105] To conform the effect of rAAV5-hIL-10/hGDNF, one of the therapeutic pharmaceutical agents of the present invention on a neurodegenerative disease, saline, rAAV5-hIL-10, rAAV5-hGDNF and rAAV5-hIL-10/hGDNF were administrated to neurodegenerative disease animal model to observe the suppression of dopaminergic neuronal cell death and the improvement of motor function.

1-1. Preparation of Animal Model

[0106] It is known that motor malfunction is showed in both animal and human with neurodegenerative disease due to changes of neurotransmitter in brain. An animal model to observe the effect of the gene therapeutic pharmaceutical agent of the present invention on ameliorating neurodegenerative diseases was induced in C57BL/6 mouse. To produce a neurodegenerative disease mouse model, 6-hydroxydopamine (6-OHDA), which is known to induce dopaminergic neuronal cell death and abnormal excitatory neurotransmission in the subthalamic nucleus (STN), was injected into the left striatum (ST) of the mouse. A specific method for inducing a neurodegenerative disease mouse model is as follows.

[0107] The mice were anesthetized by intraperitoneal administration of an Alfaxan-sedator mixture, the head of the anesthetized animal was fixed using the nose clip and ear bars of a stereotaxic instrument (RWD Life Science), and then the height of the bregma and lambda of the mouse was measured to adjust the position such that the head was horizontal. After the coordinates of a desired brain target site were confirmed based on the bregma, a hole was made using a micromotor drill (Seashin) in the skull at the confirmed position. About 2 μL of test articles (may be saline, 0.1% ascorbic acid, 6-hydroxydopamine (OHDA), or rAAV5-GFP, and may vary depending on each experiment) were administrated by using a Hamilton syringe (Hamilton) (a 10 μL syringe was used, but is not limited thereto) at a rate of 0.5 μL/min with a stereotaxic injection pump (Dongbang Hitech Inc.). After injection, the brain skin was sutured using black silk, disinfected with povidone, returned to the their homecage, and then used an infrared heater to stabilize the body temperature until it woke up from anesthesia.

1-2. Confirmation of Protective Effect of Co-Administration of hIL-10 and hGDNF on Dopaminergic Neurons in Animal Model

[0108] Specifically, in order to confirm the effect of saline, rAAV5-hIL-10, rAAV5-hGDNF and rAAV5-hIL-10/hGDNF on the suppression of dopaminergic neuronal cell death, after pretreating the mice with saline, rAAV5-hIL-10, rAAV5-hGDNF and rAAV5-hIL-10/hGDNF was pretreated and 2 weeks later, neurodegenerative disease animal model was induced by injecting 6-OHDA in mouse. 3,3'-diaminobenzidine staining (DAB staining) was performed to confirm the protective effect on dopaminergic neurons in the striatum (ST) in which the axons of the substantia nigra pars compacta (SNpc) and SNpc neuronal cell body. In the experiment, 6 weeks after administration of saline, rAAV5-hIL-10, rAAV5-hGDNF and rAAV5-hIL-10/hGDNF (4 weeks after administration of 6-OHDA), the mice were sacrificed and harvested the brain, sectioned tissue was treated with 1% H_2O_2 for 15 minutes in order to prevent non-specific color development, and then to detect tyrosine hydroxylase (TH), which is known as a representative dopaminergic neuronal marker, the sectioned tissue was treated with an anti-TH antibody and incubated at a temperature of 4° C. for 8 hours or more. The tissue was treated with a biotinylated antibody to adhere to the anti-TH antibody, an avidin-biotin enzyme conjugate was bound to the biotinylated antibody using an ABC kit, and then the bound biotin was colored brown using a DAB kit and then observed under a microscope, in bright field.

[0109] As a result, as shown in FIG. **1**A, it could be confirmed that a group pretreated with rAAV5-hIL-10/hGDNF showed a significantly lower level of dopaminergic neuronal cell death in the striatum compared to groups pretreated with saline, rAAV5-hIL-10 or rAAV5-hGDNF alone, and further, even the case where a group was pretreated with rAAV5-hIL-10/hGDNF showed a significantly lower level of dopaminergic neuronal cell death in the SNpc compared to the case where a group was not pretreated with rAAV5-hIL-10/hGDNF (pretreated with saline, rAAV5-hIL-10, or rAAV5-hGDNF alone) (FIG. **1**B).

1-3. Confirmation of Amelioration Effect of rAAV5-hIL-10/hGDNF on Hypokinesia Symptoms in Neurodegenerative Disease Animal Model

[0110] In addition, a rotation test was performed to confirm the motor function improvement effects of saline, rAAV5-hIL-10, rAAV5-hGDNF and rAAV5-hIL-10/hGDNF. For the rotation test, 0.4 mg/mL apomorphine was dissolved in saline, 4 mg/kg of apomorphine was subcutaneously administered to each neurodegenerative disease mouse model, and after 5 minutes, the direction and degree of movement of the animal were recorded using a camera for 20 minutes, and the recorded video was analyzed the amount of clockwise rotation of the animal.

[0111] As a result, the gene therapy (rAAV5-hIL-10/hGDNF) of the present invention showed an improvement in motor activity compared to saline, rAAV5-hIL-10 or rAAV5-hGDNF administrated mice, an effect of ameliorating a decline in motor function was shown (FIG. **1**C).

1-4. Confirmation of Long-Tern Therapeutic Effect of rAAV5-hIL-10/hGDNF on Neurodegenerative Disease

[0112] In order to see the long-term therapeutic efficacy of rAAV5-hIL-10/hGNDF in persistent neurodegenerative disease, one of present invention, rAAV5-hIL-10/hGDNF, one of the present invention, was administered to 6-OHDA-induced animal models one week later. Specific experimental methods are shown in the following Table 1.

TABLE 1

		Injection materials	
Group	Animals	6-OHDA lesion surgery	Test article injection
0.1% AA + Saline 6-OHDA + rAAV5-GFP 6-OHDA + rAAV5-hIL-10/hGDNF	ICR, male (8 weeks)	ST: 2 μL 0.1% AA ST: 16 μg 6-OHDA/2 μL 0.1% AA	ST: 2 μL Saline ST: rAAV5-GFP 1 × 10^{10} VG/2 μL ST: rAAV5-hIL-10/hGDNF 1 × 10^{10} VG/2 μL

[0113] As a result of confirming the effect of ameliorating an abnormal behavior of a neurodegenerative disease when rAAV5-hGDNF/hIL-10 was post-treated and conducted a rotation test in the same manner as in Example 1-3 above, as shown in FIG. **2**, it could be confirmed that even when rAAV5-hIL-10/hGDNF alone was administered, the abnormal behavior induced by 6-OHDA was ameliorated, and that such an effect was maintained for a long time even after 5 weeks had passed after the administration of rAAV5-hIL-10/hGDNF.

[0114] Through the results described above, the present inventors could confirm that when the gene therapeutic pharmaceutical agent (rAAV5-hIL-10/hGDNF) of the present invention is administered, it is possible to not only prevent but also treat the hypokinesia symptoms of a neurodegenerative disease and the preventive and therapeutic effects are maintained for a long period of time.

Example 2. Confirmation of Neurodegenerative Disease Preventive Effect when rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 are Simultaneously Administered

2-1. Confirmation of Protective Effects of rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 on Dopaminergic Neurons in Animal Model

[0115] In order to confirm the effect of rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 (simultaneously administered), which are one of the gene therapeutic pharmaceutical agents of the present invention, on the suppression of dopaminergic neuronal cell death, after pretreating the mice with rAAV5-hIL-10/hGDNF and rAAV5-hGAD65, 2 weeks later, a neurodegenerative disease was induced to mouse by administering 6-OHDA. The procedure was performed in the same manner as in Example 2-2 to confirm the protective effect on dopaminergic neurons in the striatum (ST) in which the axons of the substantia nigra pars compacta (SNpc) and SNpc neurons.

[0116] As a result, as shown in FIG. **3**A, a group pretreated with rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 showed a significantly lower level of dopaminergic neuronal cell death in the striatum compared to the no-pretreated with rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 group, and, as shown in FIG. **3**B, it could also be confirmed that the group pretreated with rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 showed a significantly lower level of dopaminergic neuronal cell death in the SNpc compared to the group treated with 6-OHDA alone.

2-2. Confirmation of Amelioration Effects of rAAV5-hIL-10/hGDNF and rAAV5-hGAD65 on Hypokinesia Symptoms in Neurodegenerative Disease Animal Model

[0117] Two weeks before inducing disease animal mouse by the method of Example 2-1 above, the mice were pretreated with rAAV5-hIL-10/hGDNF and rAAV5-hGAD65, and a rotation test (apomorphine induced rotation) and a rota-rod test were performed on the neurodegenerative disease mouse model produced by the method described above. Specific experimental methods are shown in the following Table 2.

TABLE 2

Group	Animals	Injection materials: Test article	Injection materials: 6-OHDA lesion surgery
Saline + 0.1% AA	C57BL/6, male (9 weeks)	SN: 2 μL Saline, ST: 2 μL Saline	ST: 2 μL 0.1% ascorbic acid (AA)
Saline + 6-OHDA		SN: 2 μL Saline, ST: 2 μL Saline	ST: 18 μg 6-OHDA/ 2 μL 0.1% AA
rAAV5-hIL-10/hGDNF + rAAV5-hGAD65 + 6-OHDA		SN: rAAV5-hGAD65 5×10^{10} VG/2 μL, ST: rAAV5-hGDNF/hIL-10 5×10^{10} VG/2 μL	

① Rotation Test Results

[0118] The rotation test was performed in the same manner as in Example 1-3. As a result, the neurodegenerative model preinjected with the gene therapy product of the present invention showed a significant improvement in motor function compared to the saline injection group after 4 weeks of 6-OHDA injection. (FIG. **4**A). Also, as shown in FIG. **4**B, it was found that there was a significant therapeutic effect on hypokinesia in 6-OHDA induced neurodegenerative model with pretreatment of therapeutic gene therapy compared to the saline injected group even after 10 weeks of neurodegenerative modeling. (FIG. **4**B).

[0119] Through the results described above, the present inventors could confirm that when the gene therapeutic pharmaceutical agent (simultaneous administration of rAAV5-hIL-10/hGDNF and rAAV5-hGAD65) of the present invention is administered, it is possible to prevent the hypokinesia symptoms of the neurodegenerative disease and the preventive effect is maintained for a long period of time.

② Rota-Rod Test Results

[0120] In the rota-rod test, the mice were placed on a rota-rod behavioral test apparatus cylinder and then were given time to adapt to the apparatus by rotating the disk clockwise at a speed of 5 RPM for 2 minutes. The time the mouse walked on the cylinder was measured while rotating the cylinder at various RPMs, the experiment was performed in triplicate per individual, and the average of the three measurement values was shown as the representative value of the individual.

[0121] As a result, as shown in FIG. **4**C, pre-treatment of rAAV5-hIL-10/hGDNF and rAAV5-hGAD65, one of the present inventions, showed motor improvement at a level similar to that of sham control groups which only 0.1% ascorbic acid administrated in 2 weeks, at which time point the 6-OHDA injection significantly reduced motor function.

Example 3. Confirmation of Effect of Simultaneous Administration of hIL-10 and hGDNF Supernatant or hIL-10, hGDNF, and hGAD65 Supernatant Produced by Each Therapeutic Gene Plasmid Transfection on Suppression of Neuronal Cell Death Caused by Oxidative Stress

[0122] It was aimed to see the anti-apoptotic effect of neuronal cell caused by oxidative stress in co-administration of hIL-10 and hGDNF supernatant or simultaneous administration of hIL-10, hGDNF, and hGAD65 supernatant. The combination supernatant was produced by each therapeutic gene plasmid transfection. Cell viability was observed after inducing neuronal cell death by applying H2O2 or 6-OHDA, and hIL-10 and hGDNF supernatant or simultaneous administration of hIL-10, hGDNF, and hGAD65 supernatant produced by each therapeutic gene plasmid transfection was pre-treated.

[0123] As a result, as can be seen in FIGS. **5**A and **5**B, it was confirmed that both co-administration of hIL-10 and hGDNF supernatant produced by each therapeutic gene plasmid transfection or simultaneous administration of hIL-10, hGDNF and hGAD65 supernatant produced by each therapeutic gene plasmid transfection suppressed neuronal cell death.

[0124] Through these results, it could be seen that not only the genes of hIL-10 and hGDNF, but also their respective proteins have preventive or therapeutic effects on neurodegenerative diseases, and that they are effective against various neurodegenerative diseases including Parkinson's disease because they have a neuroprotective effect regardless of the cause of neuronal cell death.

[0125] The above-described description of the present invention is provided for illustrative purposes, and those skilled in the art to which the present invention pertains will understand that the present invention can be easily modified into other specific forms without changing the technical spirit or essential features of the present invention. Therefore, it should be understood that the above-described embodiments are only exemplary in all aspects and are not restrictive.

SEQUENCE LISTING

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<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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Arg Ala Ser Pro Gly Gln Gly Thr Gln Ser Glu Asn Ser Cys Thr His
            20                  25                  30

Phe Pro Gly Asn Leu Pro Asn Met Leu Arg Asp Leu Arg Asp Ala Phe
        35                  40                  45

Ser Arg Val Lys Thr Phe Phe Gln Met Lys Asp Gln Leu Asp Asn Leu
    50                  55                  60

Leu Leu Lys Glu Ser Leu Leu Glu Asp Phe Lys Gly Tyr Leu Gly Cys
65                  70                  75                  80

Gln Ala Leu Ser Glu Met Ile Gln Phe Tyr Leu Glu Glu Val Met Pro
                85                  90                  95

Gln Ala Glu Asn Gln Asp Pro Asp Ile Lys Ala His Val Asn Ser Leu
            100                 105                 110

Gly Glu Asn Leu Lys Thr Leu Arg Leu Arg Leu Arg Arg Cys His Arg
        115                 120                 125

Phe Leu Pro Cys Glu Asn Lys Ser Lys Ala Val Glu Gln Val Lys Asn
    130                 135                 140

Ala Phe Asn Lys Leu Gln Glu Lys Gly Ile Tyr Lys Ala Met Ser Glu
145                 150                 155                 160

Phe Asp Ile Phe Ile Asn Tyr Ile Glu Ala Tyr Met Thr Met Lys Ile
                165                 170                 175

Arg Asn


<210> SEQ ID NO 2
<211> LENGTH: 1600
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 2

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gtcctcctga ctggggtgag ggccagccca ggccagggca cccagtctga gaacagctgc     120

acccacttcc caggcaacct gcctaacatg cttcgagatc tccgagatgc cttcagcaga     180

gtgaagactt tctttcaaat gaaggatcag ctggacaact tgttgttaaa ggagtccttg     240

ctggaggact ttaagggtta cctgggttgc caagccttgt ctgagatgat ccagttttac     300
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ccctgtgaaa acaagagcaa ggccgtggag caggtgaaga atgcctttaa taagctccaa    480
gagaaaggca tctacaaagc catgagtgag tttgacatct tcatcaacta catagaagcc    540
tacatgacaa tgaagatacg aaactgagac atcagggtgg cgactctata gactctagga    600
cataaattag aggtctccaa aatcggatct ggggctctgg gatagctgac ccagcccctt    660
gagaaacctt attgtacctc tcttatagaa tatttattac ctctgatacc tcaaccccca    720
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ttttttcaat atttattatt ttcacctgtt tttaagctgt ttccataggg tgacacacta    840
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<210> SEQ ID NO 3
<211> LENGTH: 537
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: codon optimized human IL-10 sequence

<400> SEQUENCE: 3

atgcacagct cagcactgct gtgctgcctg gtcctgctga caggggtgag ggcaagccca     60
ggccagggaa cccaatctga gaacagctgc acccacttcc ctggcaatct gcctaacatg    120
ctgcgcgacc tccgagatgc cttcagcaga gtgaagactt ttttccagat gaaggatcag    180
ctggacaacc tgctgctgaa ggagtccctc ctggaggact ttaagggcta cctgggatgc    240
caggccctgt ctgagatgat ccaattctac ctggaagaag ttatgcccca ggctgagaac    300
caggacccag acattaaggc ccatgtcaac tccctggggg aaaatctgaa gaccctcagg    360
ctgcggctac ggcgctgtca ccgttttctg ccctgtgaga ataagagcaa ggctgtggag    420
caggtgaaga acgccttcaa taagctccag gagaagggta tctacaaagc gatgagtgaa    480
tttgatatct tcattaatta tatagaagct tatatgacaa tgaaaatcag aaactga      537
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<210> SEQ ID NO 4
<211> LENGTH: 185
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<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 4

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Met Lys Leu Trp Asp Val Val Ala Val Cys Leu Val Leu Leu His Thr
1               5                   10                  15

Ala Ser Ala Phe Pro Leu Pro Ala Ala Asn Met Pro Glu Asp Tyr Pro
            20                  25                  30

Asp Gln Phe Asp Asp Val Met Asp Phe Ile Gln Ala Thr Ile Lys Arg
        35                  40                  45

Leu Lys Arg Ser Pro Asp Lys Gln Met Ala Val Leu Pro Arg Arg Glu
    50                  55                  60

Arg Asn Arg Gln Ala Ala Ala Ala Asn Pro Glu Asn Ser Arg Gly Lys
65                  70                  75                  80

Gly Arg Arg Gly Gln Arg Gly Lys Asn Arg Gly Cys Val Leu Thr Ala
                85                  90                  95

Ile His Leu Asn Val Thr Asp Leu Gly Leu Gly Tyr Glu Thr Lys Glu
            100                 105                 110

Glu Leu Ile Phe Arg Tyr Cys Ser Gly Ser Cys Asp Ala Ala Glu Thr
        115                 120                 125

Thr Tyr Asp Lys Ile Leu Lys Asn Leu Ser Arg Asn Arg Arg Leu Val
    130                 135                 140

Ser Asp Lys Val Gly Gln Ala Cys Cys Arg Pro Ile Ala Phe Asp Asp
145                 150                 155                 160

Asp Leu Ser Phe Leu Asp Asp Asn Leu Val Tyr His Ile Leu Arg Lys
                165                 170                 175

His Ser Ala Lys Arg Cys Gly Cys Ile
            180                 185
```

<210> SEQ ID NO 5
<211> LENGTH: 558
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

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atgaagttat gggatgtcgt ggctgtctgc ctggtgctgc tccacaccgc gtccgccttc     60

ccgctgcccg ccgcaaatat gccagaggat tatcctgatc agttcgatga tgtcatggat    120

tttattcaag ccaccattaa aagactgaaa aggtcaccag ataaacaaat ggcagtgctt    180

cctagaagag agcggaatcg gcaggctgca gctgccaacc cagagaattc cagaggaaaa    240

ggtcggagag gccagagggg caaaaaccgg ggttgtgtct taactgcaat acatttaaat    300

gtcactgact tgggtctggg ctatgaaacc aaggaggaac tgatttttag gtactgcagc    360

ggctcttgcg atgcagctga gacaacgtac gacaaaatat tgaaaaactt atccagaaat    420

agaaggctgg tgagtgacaa agtagggcag gcatgttgca gacccatcgc ctttgatgat    480

gacctgtcgt ttttagatga taacctggtt taccatattc taagaaagca ttccgctaaa    540

aggtgtggat gtatctga                                                  558
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<210> SEQ ID NO 6
<211> LENGTH: 558
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Codon optimized human GDNF DNA

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<400> SEQUENCE: 6

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ttcattcaag ccacaatcaa gcggctgaaa cgatcaccag ataaacagat ggcagtgctt    180
cctcgccgcg agcgtaatcg gcaggctgca gcagccaatc ccgagaattc ccgaggaaaa    240
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agaaggctgg tgagtgacaa ggtaggacag gcctgttgca gacccatcgc ctttgacgac    480
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```

<210> SEQ ID NO 7
<211> LENGTH: 585
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 7

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Met Ala Ser Pro Gly Ser Gly Phe Trp Ser Phe Gly Ser Glu Asp Gly
1               5                   10                  15

Ser Gly Asp Ser Glu Asn Pro Gly Thr Ala Arg Ala Trp Cys Gln Val
            20                  25                  30

Ala Gln Lys Phe Thr Gly Gly Ile Gly Asn Lys Leu Cys Ala Leu Leu
        35                  40                  45

Tyr Gly Asp Ala Glu Lys Pro Ala Glu Ser Gly Gly Ser Gln Pro Pro
    50                  55                  60

Arg Ala Ala Ala Arg Lys Ala Ala Cys Ala Cys Asp Gln Lys Pro Cys
65                  70                  75                  80

Ser Cys Ser Lys Val Asp Val Asn Tyr Ala Phe Leu His Ala Thr Asp
                85                  90                  95

Leu Leu Pro Ala Cys Asp Gly Glu Arg Pro Thr Leu Ala Phe Leu Gln
            100                 105                 110

Asp Val Met Asn Ile Leu Leu Gln Tyr Val Val Lys Ser Phe Asp Arg
        115                 120                 125

Ser Thr Lys Val Ile Asp Phe His Tyr Pro Asn Glu Leu Leu Gln Glu
    130                 135                 140

Tyr Asn Trp Glu Leu Ala Asp Gln Pro Gln Asn Leu Glu Glu Ile Leu
145                 150                 155                 160

Met His Cys Gln Thr Thr Leu Lys Tyr Ala Ile Lys Thr Gly His Pro
                165                 170                 175

Arg Tyr Phe Asn Gln Leu Ser Thr Gly Leu Asp Met Val Gly Leu Ala
            180                 185                 190

Ala Asp Trp Leu Thr Ser Thr Ala Asn Thr Asn Met Phe Thr Tyr Glu
        195                 200                 205

Ile Ala Pro Val Phe Val Leu Leu Glu Tyr Val Thr Leu Lys Lys Met
    210                 215                 220

Arg Glu Ile Ile Gly Trp Pro Gly Gly Ser Gly Asp Gly Ile Phe Ser
225                 230                 235                 240
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```
Pro Gly Gly Ala Ile Ser Asn Met Tyr Ala Met Met Ile Ala Arg Phe
                245                 250                 255

Lys Met Phe Pro Glu Val Lys Glu Lys Gly Met Ala Ala Leu Pro Arg
            260                 265                 270

Leu Ile Ala Phe Thr Ser Glu His Ser His Phe Ser Leu Lys Lys Gly
        275                 280                 285

Ala Ala Ala Leu Gly Ile Gly Thr Asp Ser Val Ile Leu Ile Lys Cys
    290                 295                 300

Asp Glu Arg Gly Lys Met Ile Pro Ser Asp Leu Glu Arg Arg Ile Leu
305                 310                 315                 320

Glu Ala Lys Gln Lys Gly Phe Val Pro Phe Leu Val Ser Ala Thr Ala
                325                 330                 335

Gly Thr Thr Val Tyr Gly Ala Phe Asp Pro Leu Leu Ala Val Ala Asp
            340                 345                 350

Ile Cys Lys Lys Tyr Lys Ile Trp Met His Val Asp Ala Ala Trp Gly
        355                 360                 365

Gly Gly Leu Leu Met Ser Arg Lys His Lys Trp Lys Leu Ser Gly Val
    370                 375                 380

Glu Arg Ala Asn Ser Val Thr Trp Asn Pro His Lys Met Met Gly Val
385                 390                 395                 400

Pro Leu Gln Cys Ser Ala Leu Leu Val Arg Glu Glu Gly Leu Met Gln
                405                 410                 415

Asn Cys Asn Gln Met His Ala Ser Tyr Leu Phe Gln Gln Asp Lys His
            420                 425                 430

Tyr Asp Leu Ser Tyr Asp Thr Gly Asp Lys Ala Leu Gln Cys Gly Arg
        435                 440                 445

His Val Asp Val Phe Lys Leu Trp Leu Met Trp Arg Ala Lys Gly Thr
    450                 455                 460

Thr Gly Phe Glu Ala His Val Asp Lys Cys Leu Glu Leu Ala Glu Tyr
465                 470                 475                 480

Leu Tyr Asn Ile Ile Lys Asn Arg Glu Gly Tyr Glu Met Val Phe Asp
                485                 490                 495

Gly Lys Pro Gln His Thr Asn Val Cys Phe Trp Tyr Ile Pro Pro Ser
            500                 505                 510

Leu Arg Thr Leu Glu Asp Asn Glu Glu Arg Met Ser Arg Leu Ser Lys
        515                 520                 525

Val Ala Pro Val Ile Lys Ala Arg Met Met Glu Tyr Gly Thr Thr Met
    530                 535                 540

Val Ser Tyr Gln Pro Leu Gly Asp Lys Val Asn Phe Phe Arg Met Val
545                 550                 555                 560

Ile Ser Asn Pro Ala Ala Thr His Gln Asp Ile Asp Phe Leu Ile Glu
                565                 570                 575

Glu Ile Glu Arg Leu Gly Gln Asp Leu
            580                 585
```

<210> SEQ ID NO 8
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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 8

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tccctcaaat gctctggggc tctccgcgct ttcctgagtc cgggctccga ggacccttag    240

gtagtcccgg tctcttttaa agctccccgg cttccaaagg gttgccacgt ccctaaaccc    300

tgtctccagc tcgcatacac acacgcacag acacgcacgt tttctgttcc tgcgtgacac    360

ccgccctcgc cgctcggccc cgccggtccc cgcgcggtgc cctcctcccg ccacacgggc    420

acgcacgcgc gcgcagggcc aagcccgagg cagctcgccc gcagctcgca ctcgcaggcg    480

acctgctcca gtctccaaag ccgatggcat ctccgggctc tggcttttgg tctttcgggt    540

cggaagatgg ctctggggat tccgagaatc ccggcacagc gcgagcctgg tgccaagtgg    600

ctcagaagtt cacgggcggc atcggaaaca aactgtgcgc cctgctctac ggagacgccg    660

agaagccggc ggagagcggc gggagccaac ccccgcgggc cgccgcccgg aaggccgcct    720

gcgcctgcga ccagaagccc tgcagctgct ccaaagtgga tgtcaactac gcgtttctcc    780

atgcaacaga cctgctgccg gcgtgtgatg gagaaaggcc cactttggcg tttctgcaag    840

atgttatgaa cattttactt cagtatgtgg tgaaaagttt cgatagatca accaaagtga    900

ttgatttcca ttatcctaat gagcttctcc aagaatataa ttgggaattg gcagaccaac    960

cacaaaattt ggaggaaatt ttgatgcatt gccaaacaac tctaaaatat gcaattaaaa   1020

cagggcatcc tagatacttc aatcaacttt ctactggttt ggatatggtt ggattagcag   1080

cagactggct gacatcaaca gcaaatacta acatgttcac ctatgaaatt gctccagtat   1140

ttgtgctttt ggaatatgtc acactaaaga aaatgagaga aatcattggc tggccagggg   1200

gctctggcga tgggatattt tctcccggtg gcgccatatc taacatgtat gccatgatga   1260

tcgcacgctt taagatgttc ccagaagtca aggagaaagg aatggctgct cttcccaggc   1320

tcattgcctt cacgtctgaa catagtcatt tttctctcaa gaagggagct gcagccttag   1380

ggattggaac agacagcgtg attctgatta aatgtgatga gagagggaaa atgattccat   1440

ctgatcttga aagaaggatt cttgaagcca aacagaaagg gtttgttcct ttcctcgtga   1500

gtgccacagc tggaaccacc gtgtacggag catttgaccc cctcttagct gtcgctgaca   1560

tttgcaaaaa gtataagatc tggatgcatg tggatgcagc ttggggtggg ggattactga   1620

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gattgatgca gaattgcaac caaatgcatg cctcctacct ctttcagcaa gataaacatt   1800

atgacctgtc ctatgacact ggagacaagg ccttacagtg cggacgccac gttgatgttt   1860

ttaaactatg gctgatgtgg agggcaaagg ggactaccgg gtttgaagcg catgttgata   1920

aatgtttgga gttggcagag tatttataca acatcataaa aaaccgagaa ggatatgaga   1980

tggtgtttga tgggaagcct cagcacacaa atgtctgctt ctggtacatt cctccaagct   2040

tgcgtactct ggaagacaat gaagagagaa tgagtcgcct ctcgaaggtg gctccagtga   2100

ttaaagccag aatgatggag tatggaacca caatggtcag ctaccaaccc ttgggagaca   2160

aggtcaattt cttccgcatg gtcatctcaa acccagcggc aactcaccaa gacattgact   2220

tcctgattga agaaatagaa cgccttggac aagatttata ataaccttgc tcaccaagct   2280

gttccacttc tctagagaac atgccctcag ctaagccccc tactgagaaa cttcctttga   2340
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gaattgtgcg acttcacaaa atgcaaggtg aacaccactt tgtctctgag aacagacgtt   2400

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agtagttgct actcttcaga atatggacaa agaaggcaca ggtgtaaata tagtagcagg   2520

atgaggaacc tcaaactggg tatcattttg cacgtgctct tctgttctca aatgctaaat   2580

gcaaacactg tgtatttatt agttaggtgt gccaaactac cgttcccaaa ttggtgtttc   2640

tgaatgacat caacattccc ccaacattac tccattacta aagacagaaa aaaataaaaa   2700

cataaaatat acaaacatgt ggcaacctgt tcttcctacc aaatataaac ttgtgtatga   2760

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tctc                                                                2824


<210> SEQ ID NO 9
<211> LENGTH: 594
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 9

Met Ala Ser Ser Thr Pro Ser Ser Ser Ala Thr Ser Ser Asn Ala Gly
1               5                   10                  15

Ala Asp Pro Asn Thr Thr Asn Leu Arg Pro Thr Thr Tyr Asp Thr Trp
            20                  25                  30

Cys Gly Val Ala His Gly Cys Thr Arg Lys Leu Gly Leu Lys Ile Cys
        35                  40                  45

Gly Phe Leu Gln Arg Thr Asn Ser Leu Glu Glu Lys Ser Arg Leu Val
    50                  55                  60

Ser Ala Phe Lys Glu Arg Gln Ser Ser Lys Asn Leu Leu Ser Cys Glu
65                  70                  75                  80

Asn Ser Asp Arg Asp Ala Arg Phe Arg Arg Thr Glu Thr Asp Phe Ser
                85                  90                  95

Asn Leu Phe Ala Arg Asp Leu Leu Pro Ala Lys Asn Gly Glu Glu Gln
            100                 105                 110

Thr Val Gln Phe Leu Leu Glu Val Val Asp Ile Leu Leu Asn Tyr Val
        115                 120                 125

Arg Lys Thr Phe Asp Arg Ser Thr Lys Val Leu Asp Phe His His Pro
    130                 135                 140

His Gln Leu Leu Glu Gly Met Glu Gly Phe Asn Leu Glu Leu Ser Asp
145                 150                 155                 160

His Pro Glu Ser Leu Glu Gln Ile Leu Val Asp Cys Arg Asp Thr Leu
                165                 170                 175

Lys Tyr Gly Val Arg Thr Gly His Pro Arg Phe Phe Asn Gln Leu Ser
            180                 185                 190

Thr Gly Leu Asp Ile Ile Gly Leu Ala Gly Glu Trp Leu Thr Ser Thr
        195                 200                 205

Ala Asn Thr Asn Met Phe Thr Tyr Glu Ile Ala Pro Val Phe Val Leu
    210                 215                 220

Met Glu Gln Ile Thr Leu Lys Lys Met Arg Glu Ile Val Gly Trp Ser
225                 230                 235                 240

Ser Lys Asp Gly Asp Gly Ile Phe Ser Pro Gly Gly Ala Ile Ser Asn
                245                 250                 255

Met Tyr Ser Ile Met Ala Ala Arg Tyr Lys Tyr Phe Pro Glu Val Lys
            260                 265                 270
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-continued

Thr Lys Gly Met Ala Ala Val Pro Lys Leu Val Leu Phe Thr Ser Glu
        275   280   285

Gln Ser His Tyr Ser Ile Lys Lys Ala Gly Ala Ala Leu Gly Phe Gly
    290   295   300

Thr Asp Asn Val Ile Leu Ile Lys Cys Asn Glu Arg Gly Lys Ile Ile
305   310   315   320

Pro Ala Asp Phe Glu Ala Lys Ile Leu Glu Ala Lys Gln Lys Gly Tyr
                325   330   335

Val Pro Phe Tyr Val Asn Ala Thr Ala Gly Thr Thr Val Tyr Gly Ala
            340   345   350

Phe Asp Pro Ile Gln Glu Ile Ala Asp Ile Cys Glu Lys Tyr Asn Leu
        355   360   365

Trp Leu His Val Asp Ala Ala Trp Gly Gly Gly Leu Leu Met Ser Arg
    370   375   380

Lys His Arg His Lys Leu Asn Gly Ile Glu Arg Ala Asn Ser Val Thr
385   390   395   400

Trp Asn Pro His Lys Met Met Gly Val Leu Leu Gln Cys Ser Ala Ile
                405   410   415

Leu Val Lys Glu Lys Gly Ile Leu Gln Gly Cys Asn Gln Met Cys Ala
            420   425   430

Gly Tyr Leu Phe Gln Pro Asp Lys Gln Tyr Asp Val Ser Tyr Asp Thr
        435   440   445

Gly Asp Lys Ala Ile Gln Cys Gly Arg His Val Asp Ile Phe Lys Phe
    450   455   460

Trp Leu Met Trp Lys Ala Lys Gly Thr Val Gly Phe Glu Asn Gln Ile
465   470   475   480

Asn Lys Cys Leu Glu Leu Ala Glu Tyr Leu Tyr Ala Lys Ile Lys Asn
                485   490   495

Arg Glu Glu Phe Glu Met Val Phe Asn Gly Glu Pro Glu His Thr Asn
            500   505   510

Val Cys Phe Trp Tyr Ile Pro Gln Ser Leu Arg Gly Val Pro Asp Ser
        515   520   525

Pro Gln Arg Arg Glu Lys Leu His Lys Val Ala Pro Lys Ile Lys Ala
    530   535   540

Leu Met Met Glu Ser Gly Thr Thr Met Val Gly Tyr Gln Pro Gln Gly
545   550   555   560

Asp Lys Ala Asn Phe Phe Arg Met Val Ile Ser Asn Pro Ala Ala Thr
                565   570   575

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Asp Leu

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<212> TYPE: DNA
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gagatctgct tccggctaag aacggtgagg agcaaaccgt gcaattcctc ctggaagtgg 360

tggacatact cctcaactat gtccgcaaga catttgatcg ctccaccaag gtgctggact 420

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accccgagtc cctggagcag atcctggttg actgcagaga caccttgaag tatggggttc 540

gcacaggtca tcctcgattt ttcaaccagc tctccactgg attggatatt attggcctag 600

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gcctcagggg tgtgccagac agccctcaac gacgggaaaa gctacacaag gtggctccaa 1620

aaatcaaagc cctgatgatg gagtcaggta cgaccatggt tggctaccag ccccaagggg 1680

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(F-JDK)

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(R-JDK)

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<210> SEQ ID NO 14
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<223> OTHER INFORMATION: Reverse primer for amplifying SV40pA promoter
      (R-SV40pA)

<400> SEQUENCE: 14

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<210> SEQ ID NO 15
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<210> SEQ ID NO 16
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1. A pharmaceutical composition for preventing or treating a degenerative brain disease, comprising two or more selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

2. The pharmaceutical composition of claim **1**, comprising the IL-10 protein or the gene encoding the same and the GDNF protein or the gene encoding the same.

3. The pharmaceutical composition of claim **2**, further comprising the GAD protein or the gene encoding the same.

4. The pharmaceutical composition of claim **1**, wherein the gene is comprised in a vector.

5. The pharmaceutical composition of claim **4**, wherein the vector is a viral or non-viral vector.

6. The pharmaceutical composition of claim **5**, wherein the viral vector is one or more selected from the group consisting of adenoviruses, adeno-associated viruses (AAV), herpes simplex virus, lentiviruses, retroviruses, cytomegalovirus, baculoviruses, and poxviruses.

7. The pharmaceutical composition of claim **5**, wherein the non-viral vector is one or more selected from the group consisting of plasmids, liposomes, cationic polymers, micelles, emulsions, and solid lipid nanoparticles.

8. The pharmaceutical composition of claim **1**, wherein the IL-10 protein is an amino acid sequence represented by SEQ ID NO: 1.

9. The pharmaceutical composition of claim **1**, wherein the IL-10 encoding gene is a base sequence represented by SEQ ID NO: 2 or SEQ ID NO: 3.

10. The pharmaceutical composition of claim **1**, wherein the GDNF protein is an amino acid sequence represented by SEQ ID NO: 4.

11. The pharmaceutical composition of claim **1**, wherein the GDNF protein encoding gene is a base sequence represented by SEQ ID NO: 5 or SEQ ID NO: 6.

12. The pharmaceutical composition of claim **1**, wherein the GAD is GAD 65 or GAD 67.

13. The pharmaceutical composition of claim **1**, wherein the GAD protein is an amino acid sequence represented by SEQ ID NO: 7 or SEQ ID NO: 9.

14. The pharmaceutical composition of claim **1**, wherein the GAD protein encoding gene is a base sequence represented by SEQ ID NO: 8 or SEQ ID NO: 10.

15. The pharmaceutical composition of claim **1**, wherein the degenerative brain disease is one or more selected from the group consisting of Parkinson's disease, dementia, Alzheimer's disease, Huntington's disease, frontotemporal lobar degeneration, dementia with Lewy bodies, vascular dementia, corticobasal degeneration, multiple system atrophy, progressive supranuclear palsy, primary lateral sclerosis, spinal muscular atrophy, stroke, cerebral infarction, head trauma, spinal cord injury, cerebral arteriosclerosis, Lou Gehrig's disease, and multiple sclerosis disease.

16. The pharmaceutical composition of claim **1**, further comprising a physiologically acceptable carrier.

17. The pharmaceutical composition of claim **1**, wherein the pharmaceutical composition is an injection.

18. The pharmaceutical composition of claim **1**, wherein the pharmaceutical composition has an effect of ameliorating motor abnormalities caused by neuronal cell death.

19. The pharmaceutical composition of claim **18**, wherein the neurons are motor neurons, sensory neurons or interneurons.

20. The pharmaceutical composition of claim **1**, wherein the pharmaceutical composition has an effect of improving cognitive ability and memory.

21. A health functional food for preventing or ameliorating a degenerative brain disease, comprising two or more selected from the group consisting of an interleukin (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same.

22. A method for preventing or treating a degenerative brain disease, the method comprising administering two or more selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same to an individual in need thereof.

23. A use of two or more selected from the group consisting of an interleukin 10 (IL-10) protein or a gene encoding the same, a glial cell line-derived neurotrophic factor (GDNF) protein or a gene encoding the same, and a glutamate decarboxylase (GAD) protein or a gene encoding the same for preparing a preventive or therapeutic agent for degenerative brain diseases.

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